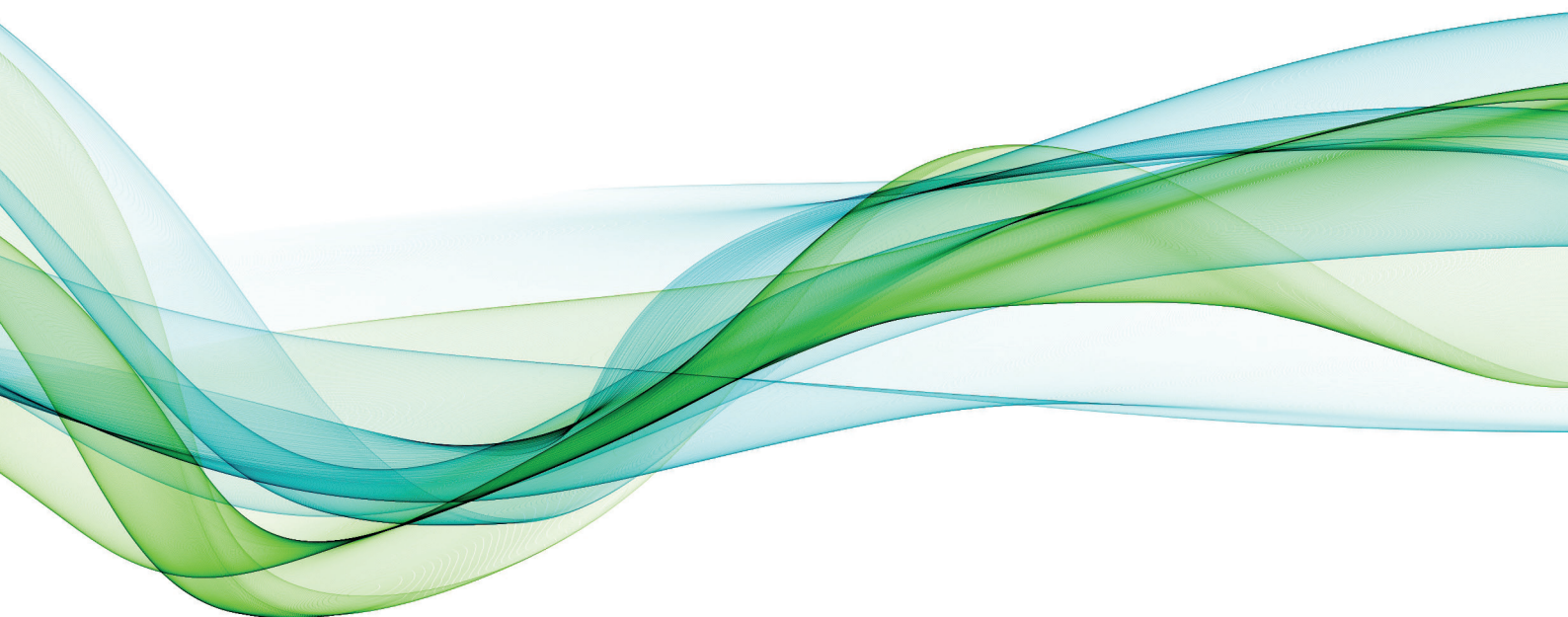


D1/A2

Technical Brochure



Advances in DGA interpretation

Reference: 771



July 2019

Advances in DGA interpretation

JWG D1/A2.47

Members

M.DUVAL , Convenor	CA
E.ALZIEU	FR
M. BANOVIC	HR
S. BHUMIWAT	NZ
P.BOMAN	US
T.BUCHACZ	PL
A.M. DALE	NO
B. DJURIK	RS
A FIELDSEND-ROXBOROUGH	GB
M.GRISARU	IL
T.HEIZMANN	CH
S. LEIVO	FI
M.A.MARTINS	PT
B.NEMETH	HU
A. DEPABLO	ES
C.PERRIER	FR
J.RASCO	US
F.SCATIGGIO	IT
H.D.SEO	KR
M.SZEBENI	HU
J.WANG	CH

I.BOCSI, Secretary	HU
O.AMIROUCHE	IT
C.BEAUCHEMIN	CA
G.J.P..DEBIJL	NL
J. BUCHACZ	PL
A. CONSTANT	FR
S.DORIEUX	FR
S.EECKOUDT	BE
R.FROETSCHER	DE
A.HADZI-SKERLEV	HR
P.KUANSATIT	TH
J.LUKIC	RS
C. MICHELLON	FR
A.NUNEZ	US
A.PEIXOTO	PT
N.PERJANIK	US
S. RYDER	GB
R.SCHNEIDER	CH
S.SPREMIC	RS
J. WALKER	FR

Copyright © 2019

"All rights to this Technical Brochure are retained by CIGRE. It is strictly prohibited to reproduce or provide this publication in any form or by any means to any third party. Only CIGRE Collective Members companies are allowed to store their copy on their internal intranet or other company network provided access is restricted to their own employees. No part of this publication may be reproduced or utilized without permission from CIGRE".

Disclaimer notice

"CIGRE gives no warranty or assurance about the contents of this publication, nor does it accept any responsibility, as to the accuracy or exhaustiveness of the information. All implied warranties and conditions are excluded to the maximum extent permitted by law".



ISBN : 978-2-85873-473-3

ISBN : 978-2-85873-473-3

Executive summary

This Technical Brochure is a continuation of the work done by CIGRE TF D1.01/ A2.11 and WG D1.32, published in TB 296 [B1] and in 2010 in TB443 [B2], respectively, on:

- The typical and pre-failure gas concentration values and rates of gas increase observed in power transformers of different types and ages (introduced in IEC 60599) [B3] and on the corresponding oil sampling intervals required.
- The identification with Triangle 4 of the stray gassing of mineral oils in service and in the laboratory, and with Triangles 3 and 6 for non-mineral oils.
- The formation of carbon oxides in transformers.
- The formation of gases from partial discharges of the corona-type PD and of the sparking-type D1.
- The application of DGA to on-load tap-changers of the compartment-type and in-tank-type, using Triangles 2.

This Technical Brochure describes:

- The different types and sub-types of faults detectable with Triangles 1, 4, 5 and Pentagons 1,2.
- The occurrence of these faults in transformers, using the DGA database of the WG, containing more than 330,000 DGA results.
- The faults of lesser concern in transformers in service (stray gassing of oil S, partial discharges of the corona-type PD, overheating $O < 250^{\circ}\text{C}$, high temperature faults in oil only T3-H, arcing faults D1 in oil only).
- The faults potentially more dangerous (carbonization of paper C, arcing in paper). Among faults C, those in leads and between turns are the least and most dangerous, respectively.
- How typical and pre-failure values are affected by these different types of faults, using the DGA database of the working group (values higher for faults of lesser concern, much lower for faults C between turns).
- How they are affected by the oxygen to nitrogen ratio, and by data from on-line gas monitors (values ten times higher).
- The typical values observed in wind-farm transformers and bushings (much higher than in power transformers).
- How to use carbon oxide gases to confirm faults in paper or in oil.
- The usual and unusual stray gassing of mineral and non-mineral oils.
- The effect of mixtures of faults on fault identification.
- How to distinguish faulty on-load tap-changers from normally operating ones with Triangles 2, particularly those of the in-tank-type.
- The evolution of normal gas formation in in-tank-types with years in service.
- The use of C_3 hydrocarbons for fault identification in transformers and on-load tap-changers.

Contents

Executive summary	5
1. Introduction.....	11
2. Faults and stresses detectable with DGA	13
2.1 DGA methods used to identify faults or stresses.....	13
2.2 Minimum gas concentration for using DGA fault or stress identification methods	13
2.3 Types of faults or stresses identifiable by DGA	13
3. How to detect transformer faults and stresses using DGA.....	17
3.1 DGA results available from the transformer	17
3.2 Stray gassing of oil S	17
3.2.1 Stray gassing in wind-farm transformers	18
3.2.2 Stray gassing in bushings.....	18
3.3 Partial discharges of the corona-type PD.....	18
3.4 Overheating O	19
3.5 Low temperature fault in paper (130-180°C).....	21
3.6 Fault T3-H (T3 in oil only).....	21
3.7 Fault D1 in oil	22
3.8 Carbonization of paper C in leads.....	24
3.9 Carbonization of paper C on windings	25
3.10 Carbonization of paper C between turns.....	27
3.11 Fault D1 in paper	29
3.12 Fault D2 in windings.....	30
4. How to detect faults in on-load tap- changers with DGA	33
4.1 Duval triangles 2 for DGA in on-load tap-changers.....	33
4.2 Normal and faulty operation of in-tank types.....	34
4.3 Factors affecting normal gas formation of in-tank types.....	36
5. Conclusions	39
APPENDIX A. Definitions, abbreviations and symbols	41
A.1. Terms and definitions	41
APPENDIX B. Links and references	43
Trademarked products.....	44
Acknowledgements.....	44
APPENDIX C. Calculations on the DGA database of WG D1/A2.47	45
C.1. DGA database of the WG	45
C.1.1. Individual DGA databases	45
C.1.2. Representativity of the DGA database of the WG	45
C.2. Identification by DGA of faults or stresses in the DGA database of the WG	46
C.2.1. Separation into sub-databases of faults or stresses	46
C.2.2. Possible misidentifications of faults or stresses.....	47
C.2.3. Occurrence of faults or stresses in transformers	48
C.2.4. Typical gas concentrations vs types of faults or stresses in transformers.....	48
C.3. DGA occurrence and severity of faults and stresses in the dga database of the WG.....	48
C.3.1. Occurrence of types of faults or stresses identified by DGA.....	48
C.3.2. Severity of types of faults or stresses	49
C.4. Typical gas concentration levels in the DGA database of the WG versus type of fault.....	49
C.4.1. Typical concentration levels.....	49
C.4.2. Typical concentration levels versus type of fault.....	50
C.5. Gas concentration levels above typical values	51
C.5.1. Gas concentration levels above typical values at CIGRE and IEC	51
C.5.2. Gas concentration levels above typical values	52
C.5.3. Gas Concentration Levels Above Typical Values versus Type of Fault.....	52

C.6.	Gas concentration levels above typical values versus location of fault	52
C.7.	Gassing rate levels above typical values	53
C.8.	Recommendations based on gas levels, type and location of faults.....	54
APPENDIX D. Typical concentration levels and faults in wind-farm transformers and bushings		55
D.1.	Wind-farm transformers.....	55
D.2.	Bushings	56
APPENDIX E. Use of carbon monoxide and carbon dioxide to detect faults in paper insulation		57
E.1.	Use of carbon monoxide and furans for detecting carbonization of paper.....	57
E.2.	Use of carbon dioxide and furans to detect low temperature faults (> 130°C) in paper.....	58
APPENDIX F. Influence of the oxygen/nitrogen ratio on typical values.....		59
F.1.	Use of the oxygen/nitrogen ratio.....	59
F.2.	Influence of the oxygen/nitrogen ratio on typical values in the DGA database of the WG 57	59
F.3.	Influence of the oxygen/nitrogen ratio on the occurrence of faults or stresses in the DGA database of the WG	59
APPENDIX G. Stray gassing of oil		61
G.1.	Definition of stray gassing of oil	61
G.2.	Detection and significance of stray gassing in transformers.....	61
G.3.	“Usual” stray gassing of mineral oil.....	61
G.4.	“Unusual” stray gassing of mineral oil	62
G.5.	Stray gassing tests on mineral oils by the WG	62
G.6.	Stray gassing of non-mineral oils in pentagons 3.....	64
G.7.	Stray gassing of oil in international standards.....	66
APPENDIX H. How to draw Duval triangles and pentagons		67
H.1.	Duval triangle 1 for mineral oils	67
H.2.	Triangles 4 and 5 for mineral oils.....	67
H.3.	Duval pentagons 1 and 2 for mineral oils.....	68
H.4.	When to use the triangles and pentagons?	69
H.5.	Numerical values for the triangles and pentagons.....	69
H.6.	Detection of multiple faults or stresses in mineral oils.....	70
H.7.	Effect of multiple faults on dga interpretation methods	70
H.8.	Duval triangles 3 and pentagons 3 for non-mineral oils	71
APPENDIX I. Fault identification methods using C₃ hydrocarbon gases.....		73
I.1.	Fault identification methods for transformers	73
I.2.	PEM method for transformers	74
I.3.	PEM method for on-load tap-changers.....	75
I.3.1.	Conventional type on-load tap-changer in normal operation, with acetylene> ethylene	75
I.3.2.	Conventional type on-load tap-changers in abnormal operation, with acetylene > ethylene	75
I.3.3.	Overheating in conventional type on-load tap-changer, with ethylene > acetylene	76
I.3.4.	Conventional type on-load tap-changer in normal operation, with ethylene > acetylene	76
I.3.5.	P.E.M. method for vacuum type on-load tap-changers.....	76

Figures and Illustrations

Figure 2.1. Gas Formation Patterns vs Temperature at Point of Gas Formation	14
Figure 4.1. Duval Triangles 2 and 2a for On-Load Tap-Changers	33
Figure 4.2. Evolution of Normal Gas Formation	36
Figure 4.3. Thermodynamic Considerations vs Gas Formation in Duval Triangle 1 [B22]	37
App Figure C.1. Probability of Having a Failure-Related Event (PFS) in % vs Gas Concentrations in ppm; T = Typical value; P = Prefailure value (PF)	51
App Figure E.1. Fault with Localized Carbonization of Paper C, not Detected by CO, CO ₂	57
App Figure F.1. Ethylene versus DDF and IFT of Oil with High Oxygen/Nitrogen Ratios ²	60
App Figure H.1. Duval Triangle 1 and calculation of triangular coordinates	67
App Figure H.2. Duval Triangles 4 and 5	68
App Figure H.3. Duval Pentagons 1 and 2	68
App Figure H.4. Mixtures of Faults vs. Single Faults in Duval Triangle 1	71
App Figure H.5. Modified Duval Triangle 1 of Spremic [B36]	71
App Figure H.6. Duval Pentagons 3 for Non-Mineral Oils	72
App Figure I.1. Theoretical Thermodynamic Formation	74
App Figure I.2. The P.E.M. method	74

Tables

Table 3.1. Concentration Levels of Hydrogen for Faults S	17
Table 3.2. Example of a Fault S ²	18
Table 3.3. Examples of Faults Partial Discharges of the Corona-Type PD	19
Table 3.4. Concentration Levels of Ethane for Faults O (Overheating)	19
Table 3.5. Example 1 of a Fault O ³	20
Table 3.6. Example 2 of a Fault O in Lead ⁴	20
Table 3.7. Examples of Low Temperature Faults in Paper	21
Table 3.8. Concentration Levels of Ethylene for Faults T3-H (Hot Spot in Oil Only)	21
Table 3.9. Example 1 of a Fault T3-H in Oil ⁵	22
Table 3.10. Example 2 of a Fault T3-H in Oil ⁶	22
Table 3.11. Concentration Levels of Acetylene for Faults D1 in Oil	22
Table 3.12. Example 1 of Fault D1 in Oil	23
Table 3.13. Example 2 of a Fault D1 in Oil ⁶	23
Table 3.14. Concentration Levels of Ethylene for Faults C in Leads	24
Table 3.15. Example 1 of a Fault C in Leads ⁵	24
Table 3.16. Example 2 of a Fault C in Leads ⁷	25
Table 3.17. Example 3 of a Fault C in Leads	25
Table 3.18. Concentration Levels of Ethylene for Faults C on Windings	25
Table 3.19. Example 1 of Fault C on Outside of Windings ⁸	26
Table 3.20. Example 2 of Faults C on Windings and Leads ⁵	26
Table 3.21. Example 3 of a Fault C on Windings ⁹	27
Table 3.22. Concentration Levels of Ethylene for Faults C between Turns	27
Table 3.23. Example 1 of Fault C in Winding Turn ¹⁰	28
Table 3.24. Example 2 of Fault C in Winding Turns ¹¹	28
Table 3.25. Example 3 of Fault C on Turn ¹¹	29
Table 3.26. Concentration Levels of Acetylene for Faults D1 in Paper of Windings	29
Table 3.27. Example of Sparking Partial Discharges D1 in Paper	30
Table 3.28. Example of Fault D2 Identified with Triangle 1 ⁷ and Buckling of Windings found by Inspection	31
Table 4.1. Normal and Faulty Operation of In-Tank Type	34
Table 4.2. Example 1 of Faulty On-load Tap-Changer of Class ARS	34
Table 4.3. Example 2 of Faulty On-load Tap-Changer of Class ARS	34
Table 4.4. Example 3 of On-load Tap-Changer of Class VRC	35
Table 4.5. Example 4 of Faulty On-load Tap-Changer Producing Arcing D1 in Transformer Main Tank	35

App Table C.1. Individual DGA Databases Used by WG47	46
App Table C.2. Typical Values of Gas Concentrations in the Database of the WG and in IEC 60599	46
App Table C.3. Occurrence of Types of Faults or Stresses identified by DGA	48
App Table C.4. Severity of Types of Faults or Stresses[B22]	49
App Table C.5.: Example of Calculations of Percentile Values of Gas Concentrations	50
App Table C.6. Typical Concentration Levels vs Type of Fault or Stress in the Subdatabases of the WG ^{1,14}	50
App Table C.7. CIGRE Gas Concentration Levels above Typical Values, in ppm	52
App Table C.8. Gassing Rate Levels Using Manual Oil Sampling (in ppm/month) when Faults Were Not Identified by DGA	53
App Table D.1. 90% percentiles of gas concentrations in ppm	55
App Table D.2. Relative proportion of faults, in %,	56
App Table D.3. 95% percentiles of gas concentrations in ppm	56
App Table F.1. Influence of Oxygen/Nitrogen Ratio on Typical Values in the DGA Database of the WG	59
App Table F.2. Influence of Oxygen/Nitrogen Ratio on the Occurrence of Faults in the Database of the WG	60
App Table G.1. Stray Gas Formation from Commercial Oils HV ¹⁸	63
App Table G.2. Stray Gas Formation between 70°C and 200°C with and without BHT Additive	64
App Table G.3. Stray Gassing Tests with Ester Oils in Italy ¹⁵	65
App Table G.4. Stray Gassing Tests with Ester Oils in the US ¹³	65
App Table H.1. Numerical Values for Zone boundaries in Triangles 1, 4, 5	69
App Table H.2. Numerical Values for Zone Boundaries of the Pentagons	70
App Table I.1. Diagnosis Method for C3 Hydrocarbons in Poland ¹⁹	73
App Table I.2. MSS Diagnosis Method for C3 Hydrocarbons (simplified version ²⁰)	73
App Table I.3. MSS Diagnosis Method for C3 Hydrocarbons (full version ²⁰)	73
App Table I.4. Fault Identification Using C3 Hydrocarbon Gases	75
App Table I.5. Case Studies of Conventional Type On-Load Tap-Changer in Normal Operation	75
App Table I.6. Case Studies of Conventional Type On-Load Tap-Changers in Abnormal Operation	75
App Table I.7. PEM Value and What was Found in Each Case	76
App Table I.8. Case Studies of Conventional Type On-Load Tap-Changers in Normal Condition in spite of Ethylene > Acetylene	76

1. Introduction

The scope (terms of reference) of Working Group D1/A2.47, as approved by Study Committees D1 and A2 in June 2011, was “to investigate the following items, all related to transformers and their accessories:

- case studies and suggestions for typical problems and actions initiated by DGA interpretation.
- significance of CO, CO₂ and CO₂/CO for detecting paper involvement in faults.
- review progress in the field of DGA”.

The only actions recommended in 2011 by CIGRE D1/A2 (in TB 296 [B1] and 443 [B2]) and by IEC TC10 (in IEC 60599 [B3]) were to increase sampling intervals for DGA, based on gas limits in transformers.

CIGRE/IEC and IEEE gas limits in 2011 were given for DGA results where the location of the fault (oil or paper) is not known or is uncertain. However, previous investigations by Working Group D1.32^{1,23,26} presented in Budapest in May 2011 had shown that the type of fault involved and its location have a great influence on the risk of failure of transformers.

The first two items were thus investigated by Working Group D1/A2.47 in order to propose more appropriate actions on the equipment taking into account the type and location of faults (in oil or in paper).

Particular items concerning progress in the field of DGA mentioned by former members of WG D1/A2.32 and members of Working Group D1/A2.47 were:

- DGA in on-load tap-changers of the in-tank type.
- DGA in wind farm transformers and bushings.
- DGA in ester oils
- stray gassing of oils.
- evaluation of commercial software for DGA interpretation.

Other items concerning new on-line gas monitors on the market and their accuracy as compared to laboratory results are reported in the second TB No 2 to be published by WG D1/A2.47 [B4].

Note: superscripts in this document refer to unpublished contributions by individual members indicated in the Acknowledgments section of APPENDIX B.

2. Faults and stresses detectable with DGA

2.1 DGA methods used to identify faults or stresses

The main methods used to identify faults (above typical values) or stresses (below typical values) in transformers and accessories filled with mineral oils are the Duval Triangles and Pentagons, IEC Ratios, Rogers, Dornenburg, Key Gas methods, together with dozens of other, lesser used published methods using for instance neural networks. They all primarily use hydrogen, methane, ethylene, ethane and acetylene for fault identification.

The Triangle and Pentagon methods [B5] have been used for fault identification in this TB rather than the IEC Ratio method [B3] and the other methods listed above, for the following reasons:

- the IEC method and the other two-gas ratio methods such as Rogers or Dornenburg cannot identify faults in about 15 to 20% of DGA cases, because DGA results fall outside of code zones, even when ppm values are well above typical values. The Key Gas method produces ~ 50% of wrong fault identifications when used with software [B6]. Using these methods would therefore have distorted calculations of types of faults made on the DGA database of the WG.

The Triangle and Pentagon methods do not have these limitations, always providing a fault identification (they are “closed” systems), with a high % of correct predictions since they are based on a large number of inspected cases [B6].

- the IEC method only identifies the six basic types of faults (see 2.3), not the four other sub-types of thermal faults identifiable with the Triangle and Pentagon methods (S, O, C, T3-H), which provide very useful complementary information concerning actions on the equipment (e.g., distinction between faults T3-H and C).
- algorithms were readily available for the Triangle and Pentagon methods for calculations on the DGA database of the WG, not for the other methods mentioned above.

A description of Triangles 1, 4, 5 and Pentagons 1, 2 methods for mineral oils is indicated in APPENDIX H.

2.2 Minimum gas concentration for using DGA fault or stress identification methods

The basic requirement for using DGA fault or stress identification methods (Triangles, Pentagons, Rogers, IEC Ratios, etc), is that at least one of the gas concentration values in ppm used meets the IEC requirements for accuracy (i.e, $\pm 15\%$ above 10 ppm of gas), see [B4].

Between 10 and 5 ppm, the required accuracy of IEC is $\pm 30\%$. These lower ppm values can be used for stress identification but the area of uncertainty of the DGA point should be calculated [B7] if it is close to a fault zone boundary.

If the laboratory used cannot certify that its ppm values meet the IEC requirements for accuracy, then the recommendation of IEC is that fault or stress identification methods should be used only above typical values.

2.3 Types of faults or stresses identifiable by DGA

Table 2.1 indicates the 10 types of faults (above typical values) or stresses (below typical values) identifiable by DGA [B5].

- the 6 basic types of faults or stresses in transformers defined in IEC 60599 (T3, T2, T1, D2, D1, PD) [B2] and identifiable with Triangle 1 and Pentagon 1.
- the 4 additional sub-types of thermal faults or stresses identifiable with Triangles 4, 5 and Pentagon 2 (S, O, C, T3-H).

Table 2.1. Definition of the 10 Types of Faults or Stresses Identifiable by DGA with Triangles 1, 4, 5 and Pentagons 1,2

Fault or Stress	Definition
T3	Thermal $T > 700\text{ }^{\circ}\text{C}$
T2	Thermal $300 < T < 700^{\circ}\text{C}$
T1	Thermal $< 300\text{ }^{\circ}\text{C}$
PD	Partial Discharges of the Corona Type
D1	Discharges of Low Energy
D2	Discharges of High Energy
S	Stray Gassing of Oil $< 200\text{ }^{\circ}\text{C}$
O	Overheating $< 250\text{ }^{\circ}\text{C}$
C	Possible Carbonization of Paper $> 300\text{ }^{\circ}\text{C}$
T3-H	T3 in Oil Only

Gas formation patterns vs temperature (energy content) of the fault or stress and vs. type of fault or stress are indicated in Figure 2.1 [B8]. They are the same in all types of equipment (power transformers, air breathing- or sealed-, core- or shell-type, instrument transformer of the current- or voltage-type, bushings, cables).

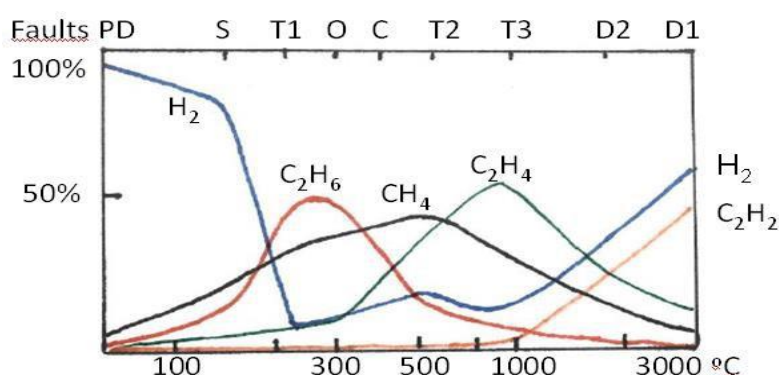


Figure 2.1. Gas Formation Patterns vs Temperature at Point of Gas Formation

Notes to Table 2.1 and Figure 2.1¹:

Note 1- according to physics, arcs in air or in liquids occur in a tiny column (path) of highly ionized gas of temperature $> 3000^{\circ}\text{C}$ and of virtually zero electrical resistance.

In case of arcs D1 of low energy in the oil of transformers, arc duration is very short, so that only a small layer of oil comes in contact with the arc path and is decomposed at its high temperature $> 3000^{\circ}\text{C}$, producing mostly acetylene and very little ethylene, on the right side of Figure 2.1.

In arcs D2 of high energy, more current flows through the arc path and arc duration is longer. This heats up a larger volume of oil through convection of cooler surrounding oil, and produces a larger gradient of temperature in the oil around the arc path (e.g., from 3000 to 500°C). As a result, the average temperature of oil in arcs D2 in transformers is lower than in arcs D1, despite their higher energy content, producing a lot of ethylene in addition to acetylene, as illustrated in Figure 2.1 and Figure 4.3.

Note 2- in terms of gas formation, there are two very different types of partial discharges, as explained in IEC 60599 [B3]:

- partial discharges of the corona-type (PD) occurring in a gas phase, e.g, in gas bubbles or voids trapped in paper as a result of poor drying or poor impregnation of oil in paper. These PDs ionize air or nitrogen in the gas phase creating a plasma of ionized oxygen and nitrogen atoms which react with the surrounding oil or cellulose, producing mainly hydrogen and appearing on the left side of Figure 2.1 as faults PD. This is the gas formation measured by the so-called “gassing tendency tests” of IEC [B9] and ASTM [B10] (see APPENDIX GG.1).
- partial discharges of the sparking type, occurring in the liquid (oil) or solid (paper) phase and which are actually small arcs, producing a lot of acetylene in addition to hydrogen and appearing on the right side of Figure 2.1 as faults D1.

Note 3- catalytic reactions of the electrolysis-type between free water and oil sampling valves made of galvanized (zinc-coated) steel can produce large quantities of hydrogen at ambient temperature or above, and appear on the leftmost side of Figure 2.1. Oil sampling valves on transformers, old or new, are today all made up of brass or stainless steel, which do not produce such catalytic reactions any more.

Note 4- for the definition of stray gassing faults S, see APPENDIX G G.1.

More than 150 inspected cases of faults in transformers filled with mineral oils have been reported by members of the WG, covering all the types of faults in Table 2.1 and Figure 2.1, practically all of them identified and predicted correctly with Triangles 1, 4, 5 and Pentagons 1, 2.

The most frequent faults reported were: faults O (50), T3-H (26), D2 (22), C (20), S (17), D1 in oil (15), D1 in paper (8), PD (4) (see Chapter 3). Mixtures of faults were also reported in 23 cases (see APPENDIX HH.6 and H.7).

For the identification of faults using the C₃ hydrocarbon gases in mineral oils see APPENDIX I.

For the identification of faults in non-mineral oils (natural and synthetic esters, silicones), only relatively minor adjustments to fault zone boundaries of the Triangles and Pentagons for mineral oil are needed, as indicated in [B2]. The Duval Triangles 3 and Pentagons 3 have been developed to detect the 6 basic types of faults and stray gassing S in non-mineral oils. They are described in APPENDIX HH.8 and G.6, respectively.

3. How to detect transformer faults and stresses using DGA

3.1 DGA results available from the transformer

When only DGA readings from on-line gas monitors of type M1 to M6 measuring one to six gases [B4] are available, full identification of the type and sub-type of fault or stress in the transformer is not possible.

In such cases, use App Table C.7 of APPENDIX C to determine gas concentration levels in the transformer (typical and above) then APPENDIX C.8 for recommendations based on these gas levels.

When laboratory DGA results or readings from on-line gas monitors of type M7 to M9 measuring 7 to 9 gases [B4] are available, full identification of the type or sub-type of fault or stress in the transformer is possible. Use APPENDIX H for fault identification then move to the corresponding type of fault or stress 3.2 to 3.12 occurring in the transformer, listed by increasing order of severity below.

3.2 Stray gassing of oil S

For definition of stray gassing of oil S, see APPENDIX G.1.

Faults or stresses S can be identified with Triangle 4 (APPENDIX H.2) and Pentagons 1 and 2 (APPENDIX H.3).

The most important gas formed with faults or stresses S is usually hydrogen (see APPENDIX G.3).

Concentration levels of hydrogen for faults stray gassing of oil S in power transformers are indicated in Table 3.1.

**Table 3.1. Concentration Levels of Hydrogen for Faults S
(Stray Gassing of Oil) in Power Transformers**

H ₂ Conc. Level	Typical	Intermediate 1	Intermediate 2	Pre-failure PF
ppm	460	800	1100	*

For definition of gas concentration levels in Table 3.1, see APPENDIXES C.4.1. and C.5.

Concentration levels for gases other than hydrogen remain the same as in App Table C.7 of APPENDIX C.

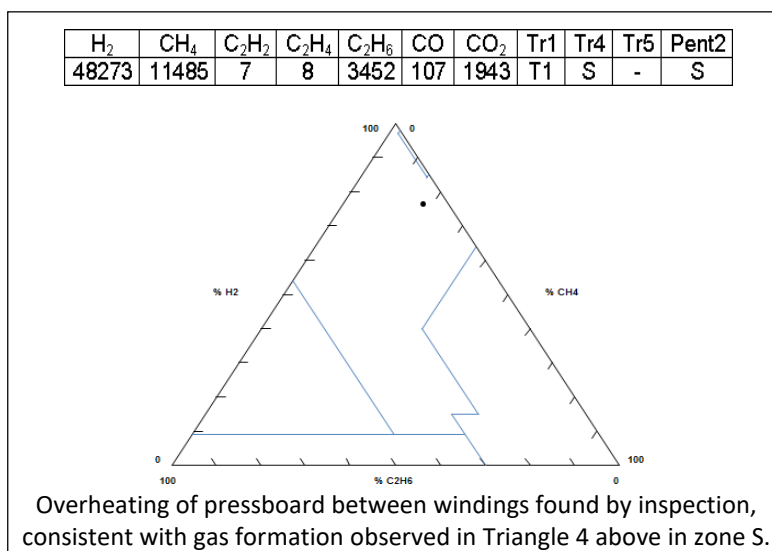
See APPENDIXES C.4.2. and C.5.3. on how concentration levels of hydrogen in case of faults S were calculated in Table 3.1.

Note *: The calculated prefailure value PF for faults S is 2840 ppm. However, more than 15 cases of faults S have been reported to the WG, with hydrogen values between 8400 and 55,000 ppm and no failure. So a PF value much higher than this calculated PF value could probably be used in Table 3.1 and intermediate values 1 and 2 recalculated accordingly. No case of failure due to faults S have been reported to the WG at any hydrogen level, high or low. Reasons for the absence of failures due to faults S are discussed in APPENDIX G.2.

For recommendations based on gas concentration levels, see APPENDIX C.8.

The occurrence of faults or stresses S in power transformers is indicated in App Table C.3 of APPENDIX C.

An example of a fault S in a sealed power transformer with coated paper is indicated in Table 3.2. For the definition of fault zones in Table 3.2, see App Figure H.2 in APPENDIX H.

Table 3.2. Example of a Fault S²

3.2.1 Stray gassing in wind-farm transformers

The occurrence of faults or stresses S in wind-farm transformers is much higher than in power transformers, as indicated in App Table D.2 of APPENDIX D.

The typical level of hydrogen in wind-farm transformers (indicated in App Table D.1 of APPENDIX D) is also much higher than in power transformers, as well as pre-failure (PF) values (tens of thousands of ppm with no failure, as reported in APPENDIX D.1). Values in Table 3.1 can therefore be significantly increased in the case of wind-farm transformers.

3.2.2 Stray gassing in bushings

The occurrence of faults or stresses S in bushings is also much higher, as well as typical values reported in App Table D.3 of APPENDIX D, and therefore pre-failure values.

As indicated in APPENDIX G.2, faults S are of relatively little concern in oil-filled electrical equipment (power transformers, wind farm transformers and bushings), even at very high levels of hydrogen, well above the calculated pre-failure (PF) value.

3.3 Partial discharges of the corona-type PD

For definition of partial discharges of the corona-type PD in transformers, see 2.3, Note 2.

Faults or stresses partial discharges of the corona-type PD can be identified with Triangle 4 (APPENDIX H.2) and Pentagons 1 and 2 (APPENDIX H.3). They are sometimes difficult to distinguish from stray gassing faults S. See APPENDIX G.2 (last para) on how to make the distinction.

The main gas formed with faults or stresses partial discharges of the corona-type PD is hydrogen.

Table 3.1 above can be used by default to determine the concentration levels of hydrogen for faults partial discharges of the corona-type PDs.

Values in Table 3.1, however, can be significantly increased in case of partial discharges of the corona-type PD, since higher levels of hydrogen tend to be produced by these faults than by faults stray gassing of oil S. The calculated typical value of H₂ in case of partial discharges of the corona-type PD is indeed in the thousands of ppm in App Table C.6, even though its exact value is a bit uncertain because of the very small number of such faults in the DGA database of the WG.

Faults partial discharges of the corona-type PD occur in only 0.04% of DGA cases in power transformers, as indicated in App Table C.3 of APPENDIX C. They occur much more frequently in wind farm transformers (see App Table D.2 of APPENDIX D), bushings (App Table D.3. of APPENDIX D), instrument transformers and cables.

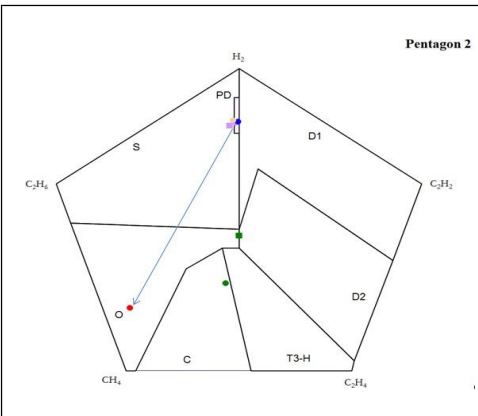
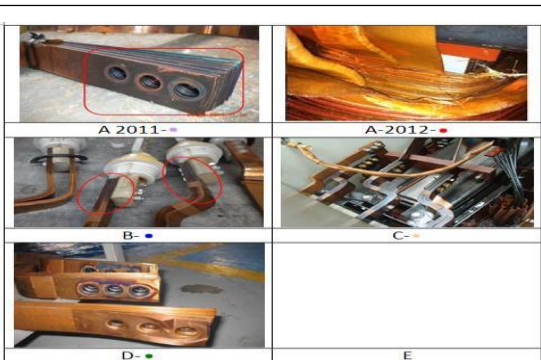
They often produce very large amounts of hydrogen without affecting the normal operation of transformers. Indeed, more than 30 such cases have been reported to the WG, with hydrogen values between 5000 and 33,000 ppm and no failure, particularly in wind farm transformers and bushings. No case of failure due to faults corona PDs have been reported to the WG at any hydrogen level, high or low, probably because they do not damage insulation.

Faults partial discharges of the corona-type PDs are of relatively little concern in electrical equipment (power, wind farm transformers and bushings, see APPENDIX D.1) as long as they do not evolve into potentially more dangerous partial discharges of the sparking type D1.

Examples of faults partial discharges of the corona-type PDs and stray gassing of oil S with high hydrogen formation and no failure¹ are indicated in Table 3.3.

Table 3.3. Examples of Faults Partial Discharges of the Corona-Type PD And Stray Gassing of Oil S in Transformers

	Date	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	Tr4	Pent2	
A	2011	10000	1000	0	0	200	S	PD	●
	2012	0	2000	0	0	700	C/O	O	●
B		15000	1500	0	0	0	PD	PD	●
C		12000	1000	0	0	500	S	S	●
D		300	150	0	75	0	S	C	●
E		6000	600	0	0	400	S	S	●

Notes: in A-2012, fault O was in windings, other cases of S or PD faults were on metal in oil.
 In D there was a small layer of paper between metal parts, hence the small fault C.
 No photo is available for case E

3.4 Overheating O

Faults overheating O < 250°C can be identified with Triangle 4 (APPENDIX H.2) and Pentagons 1 and 2 (APPENDIX H.3).

The most important gas formed in faults or stresses overheating O is ethane.

Concentration levels of ethane for faults O are indicated in Table 3.4.

Table 3.4. Concentration Levels of Ethane for Faults O (Overheating)

C ₂ H ₆ Conc. Level	Typical	Intermediate 1	Intermediate 2	Pre-failure PF
ppm	550	1000	1500	4460

For definition of gas concentration levels in Table 3.4, see APPENDIXES C.4.1 and C.5. Concentration levels for gases other than ethane remain the same as in App Table C.7 of APPENDIX C.

See APPENDIXES C.4.2 and C.5.3 on how concentration levels of ethane in case of faults O were calculated in Table 3.4.

For recommendations based on gas concentration levels see APPENDIX C.8.

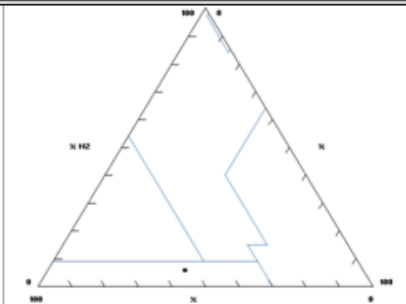
The occurrence of faults or stresses O in power transformers is indicated in App Table C.3 of APPENDIX C.

Faults O often produce large quantities of ethane without affecting normal operation. Indeed, more than seven inspected cases of faults O have been reported to the WG, with ethane values between 400 and 1200 ppm and no failure.

Example 1 of a fault O³ is indicated in Table 3.5.

Table 3.5. Example 1 of a Fault O³

Example 3: Transformer 6kV 25MVA											
H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	CO	CO ₂	Tr1	Tr4	Tr5	Pent2	
29	204	0	17	264	24	2000	T1	O	(O)	O	



Triangle 4

Overheated insulated iron sheets of lower frame found by inspection, as predicted by Triangle 4 and Pentagon 2




Example 2 of a fault O⁴ in lead of 345 kV, 460 MVA GSU transformer, with 1216 ppm ethane, 416 ppm methane, no failure is indicated in Table 3.6.

Table 3.6. Example 2 of a Fault O in Lead⁴



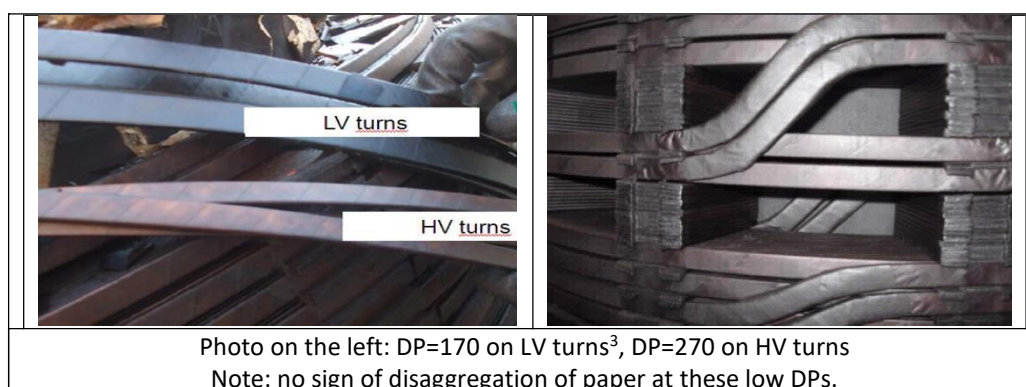
3.5 Low temperature fault in paper (130-180°C)

Most important gas formed with this type of fault is carbon dioxide, together with furans below a degree of polymerization of paper (DP) of 400.

No failures due to this type of fault have been observed, even at cumulative levels of carbon dioxide > 35,000 ppm, carbon dioxide to carbon monoxide ratios > 50, and DPs < 200 [B11]. See APPENDIX E.2 for the operation of such transformers.

Examples of low temperature faults in paper^{3,1}, no failure, is indicated in Table 3.7.

Table 3.7. Examples of Low Temperature Faults in Paper



3.6 Fault T3-H (T3 in oil only)

Faults T3-H in oil can be identified with Triangle 5 (APPENDIX H.2) and Pentagons 2 (APPENDIX H.3).

The most important gas formed is ethylene.

Concentration levels of ethylene for faults T3-H are indicated in Table 3.8.

Table 3.8. Concentration Levels of Ethylene for Faults T3-H (Hot Spot in Oil Only)

C ₂ H ₄ Conc. Level	Typical	Intermediate 1	Intermediate 2	Pre-failure PF
ppm	126	270	450	1800

For definition of gas concentration levels in Table 3.8, see APPENDICES C.4.1 and C.5.

Concentration levels for gases other than ethylene remain the same as in App Table C.7 of APPENDIX C.

See APPENDIX C.4.2 and C.5.3 on how concentration levels of ethylene in case of faults T3-H were calculated in Table 3.8.

For recommendations based on gas concentration levels see APPENDIX C.8.

Faults or stresses T3-H in oil are those occurring most frequently in power transformers, as indicated in App Table C.3 of APPENDIX C. They often produce very large quantities of ethylene without affecting the normal operation of transformers, because they do not affect the electrical insulation properties of oil and are not in paper.

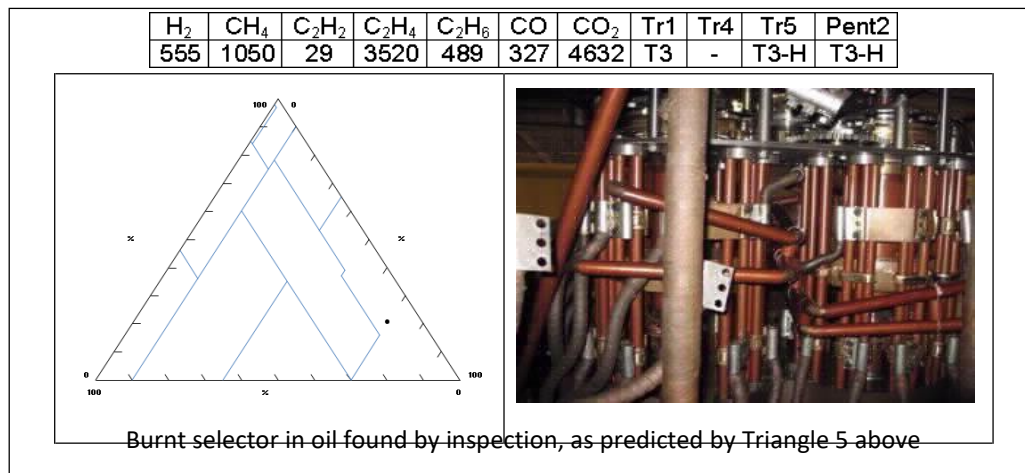
Indeed, very large quantities of ethylene (between 3500 and 440,000 ppm) have been reported to the WG in ~ 20 inspected cases of stable faults T3-H (see for instance example 2 below with more than 200,000 ppm, no failure). Typical such cases are faults in high resistance contact or connection and also in the core, like an unintentional core ground¹³.

In the very large majority of cases, faults T3-H are therefore not too much of a concern in transformers, as long as they are stable and do not evolve into more dangerous faults C or D1.

In a minority of cases, however, they may be located in a critical component (e.g., a jammed DETC) that needs to be fixed rapidly. To determine the location of faults T3-H, acoustic emission or other tests can be used (see APPENDIX C.8).

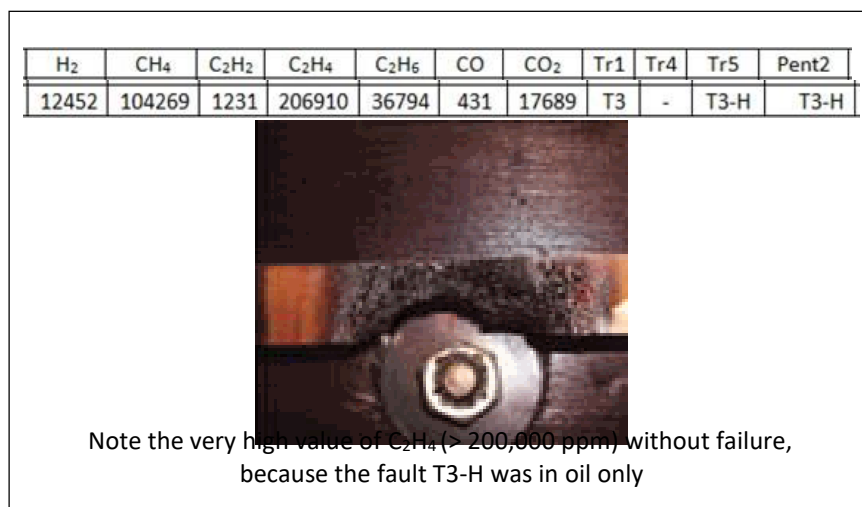
Example 1 of a fault T3-H in oil in a rectifier of 20 kV, 20 MVA is indicated in Table 3.9.

Table 3.9. Example 1 of a Fault T3-H in Oil⁵



Example 2 of a Fault T3-H in Oil⁶ is indicated in Table 3.10.

Table 3.10. Example 2 of a Fault T3-H in Oil⁶



3.7 Fault D1 in oil

Faults D1 can be identified with Triangle 1 (APPENDIX H.1) and Pentagons 1, 2 (APPENDIX H.3). Also with the IEC, Rogers and Dornenburg methods.

To determine that a fault D1 is in oil only, acoustic emission or other tests can be used (see APPENDIX C.8). Also carbon monoxide, carbon dioxide and furans (see APPENDIX E.1).

The most important gas formed is acetylene.

Concentration levels of acetylene for faults D1 in oil are indicated in Table 3.11.

Table 3.11. Concentration Levels of Acetylene for Faults D1 in Oil

C ₂ H ₂ Conc. Level	Typical	Intermediate 1	Intermediate 2	Pre-failure PF
ppm	25	60	100	1400

For definition of gas concentration levels in Table 3.11, see APPENDIXES C.4.1 and C.5. Concentration levels for gases other than acetylene remain the same as in App Table C.7 of APPENDIX C.

If the fault cannot be confirmed as being in oil only, use App Table C.7 for acetylene.

See APPENDIXES C.4.2, C.5.3 and C.6 on how concentration levels of acetylene in case of faults D1 in oil were calculated in Table 3.11.

For recommendations based on gas concentration levels see APPENDIX C.8.

Occurrence of faults or stresses D1 in power transformers is indicated in App Table C.3 of APPENDIX C.

Faults D1 in oil often produce large amounts of acetylene without affecting normal operation. Indeed, 15 inspected cases of faults D1 in oil have been reported to the WG, with acetylene values between 200 and 1400 ppm and no failure.

As soon as such faults D1 in oil evolve into a fault D2, however, immediate action is usually required.

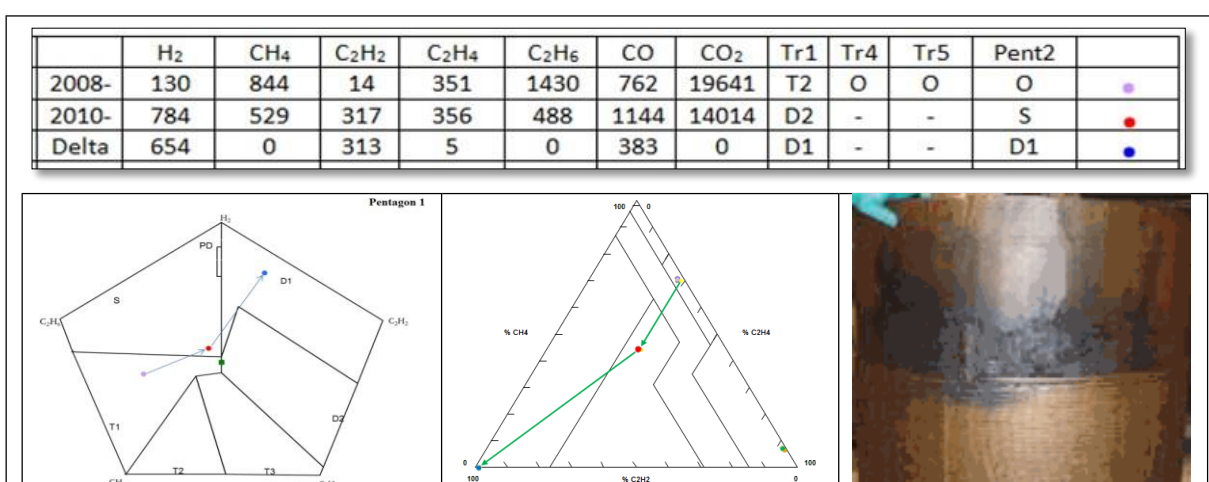
Example 1 of a tracking fault D1 in oil on a phenolic plate¹ with 480 ppm of acetylene and no failure, is indicated in Table 3.12.

Table 3.12. Example 1 of Fault D1 in Oil



Example 2 of a fault D1 in oil⁶ is indicated in Table 3.13.

Table 3.13. Example 2 of a Fault D1 in Oil⁶



Note to example 2: when moving from fault O to fault D1 in oil in Table 3.13, the DGA point appears temporarily and inappropriately in zone D2 of Triangle 1 and in zone S of Pentagon 2, because of the mixture of faults and gases (see APPENDIXES H.6 and H.7).

3.8 Carbonization of paper C in leads

Faults C (carbonization of paper) can be identified with Triangle 5 (APPENDIX H.2) and Pentagons 2 (APPENDIX H.3).

In case of faults carbonization of paper C, it is important to try to determine the location of the fault (e.g., in leads, on windings or between turns), because this will result in very different warning levels of ethylene and actions on the equipment (compare Table 3.14, Table 3.18 and Table 3.22).

To determine that a fault C is in leads, acoustic emission or other tests can be used (see APPENDIX C.8). Also carbon monoxide, carbon dioxide and furans (see APPENDIX E.1).

The most important gas formed is ethylene.

Concentration levels of ethylene for faults C in leads are indicated in Table 3.14.

Table 3.14. Concentration Levels of Ethylene for Faults C in Leads

C ₂ H ₄ Conc. Level	Typical	Intermediate 1	Intermediate 2	Pre-failure PF
ppm	200	440	700	2900

For definition of gas concentration levels in Table 3.14, see APPENDIX C.4.1 and C.5.

Concentration levels for gases other than ethylene remain the same as in App Table C.7 of APPENDIX C.

If the fault cannot be confirmed to be in leads, use App Table C.7 for ethylene.

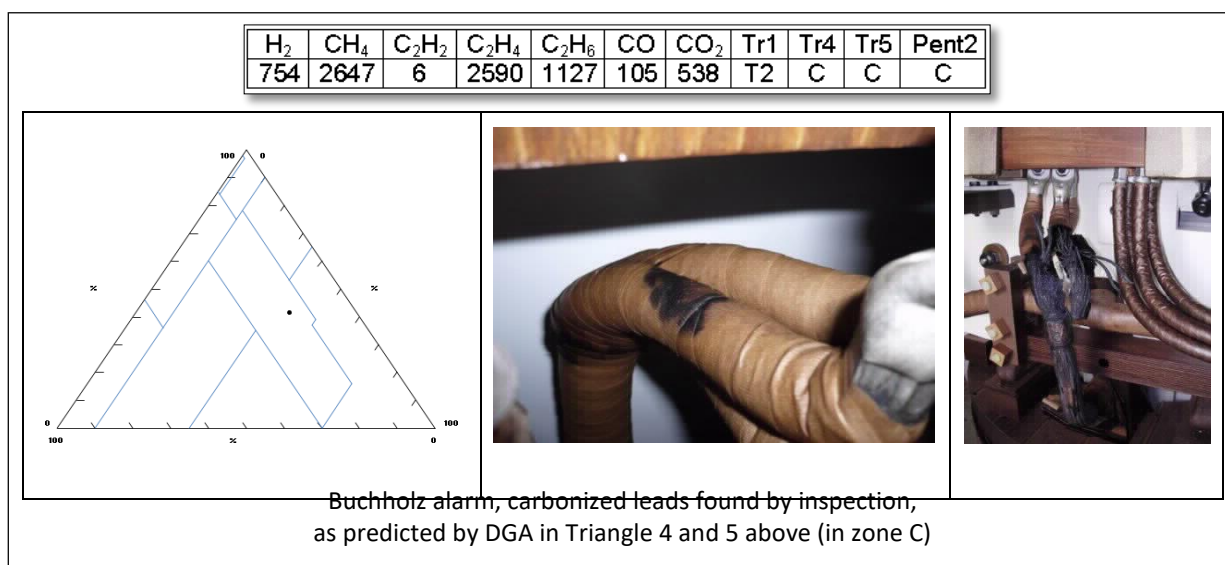
See APPENDIXES C.4.2, C.5.3 and C.6 on how concentration levels of ethylene in case of faults C in leads were calculated in Table 3.14.

For recommendations based on gas concentration levels see APPENDIX C.8.

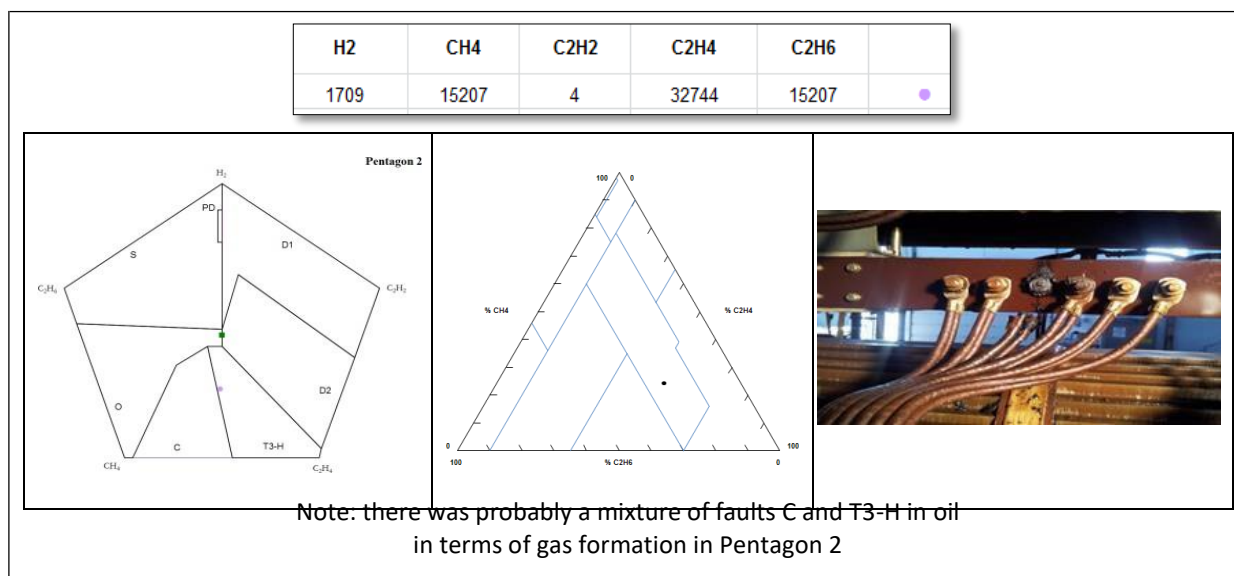
Faults or stresses C in leads are the least dangerous of faults C (carbonization of paper), because leads are usually located outside of windings and subjected to a lower voltage. Indeed, ~ 15 inspected cases have been reported to the WG, with ethylene values between 4800 ppm and 38,000 ppm and no failure.

Example 1 of a fault C in leads in a transformer of 24 kV, 40 MVA⁵ is indicated in Table 3.15.

Table 3.15. Example 1 of a Fault C in Leads⁵

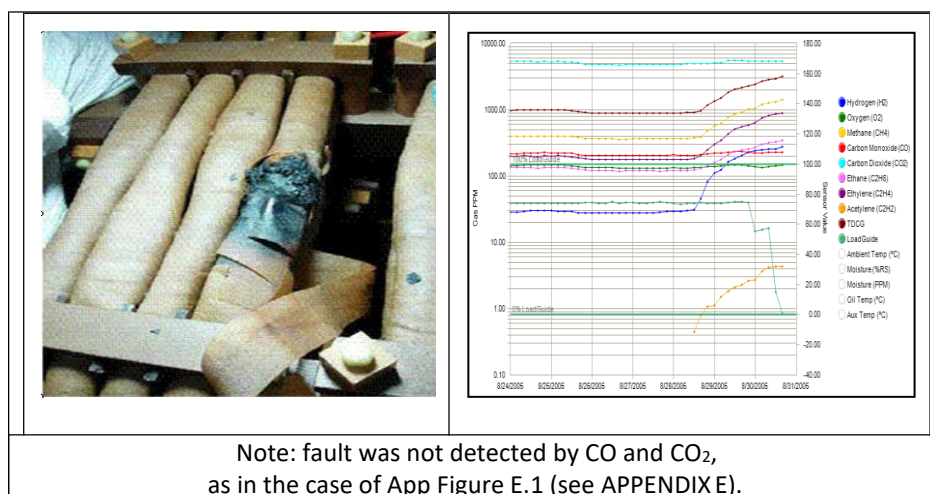


Example 2 of a fault C in leads⁷ is indicated in Table 3.16.

Table 3.16. Example 2 of a Fault C in Leads⁷

Example 3 of a fault C in leads is indicated in Table 3.17 [B8].

Table 3.17. Example 3 of a Fault C in Leads



3.9 Carbonization of paper C on windings

Faults C (carbonization of paper) can be identified with Triangle 5 (APPENDIX H.2) and Pentagons 2 (APPENDIX H.3).

To determine that a fault carbonization of paper C is on windings, acoustic emission or other tests can be used (see APPENDIX C.8). Also carbon monoxide, carbon dioxide and furans (see APPENDIX E.1).

The most important gas formed is ethylene.

Concentration levels of ethylene for faults carbonization of paper C on windings are indicated in Table 3.18.

Table 3.18. Concentration Levels of Ethylene for Faults C on Windings

C ₂ H ₄ Conc. Level	Typical	Intermediate 1	Intermediate 2	Pre-failure PF
ppm	68	150	240	970

For definition of gas concentration levels in Table 3.18, see APPENDIXES C.4.1 and C.5.

Concentration levels for gases other than ethylene remain the same as in App Table C.7 of APPENDIX C.

If the fault cannot be confirmed as being on windings, use App Table C.7 for ethylene.

See APPENDICES C.4.2, C.5.3 and C.6 on how concentration levels of ethylene in case of faults C on windings were calculated in Table 3.18.

For recommendations based on gas concentration levels see APPENDIX C.8.

Faults C (carbonization of paper) on the outside of windings do not always result in failure, as they may be subjected to a lower voltage than between turns (see example 1).

Example 1 of a fault C on the outside of tertiary windings of a 230 kV, 336 MVA transformer, with 6035 ppm ethylene, 5126 ppm methane, localized by acoustic tests, no failure⁸, is indicated in Table 3.19.

Table 3.19. Example 1 of Fault C on Outside of Windings⁸



Example 2 of faults C on windings and leads⁵ is indicated in Table 3.20.

Table 3.20. Example 2 of Faults C on Windings and Leads⁵

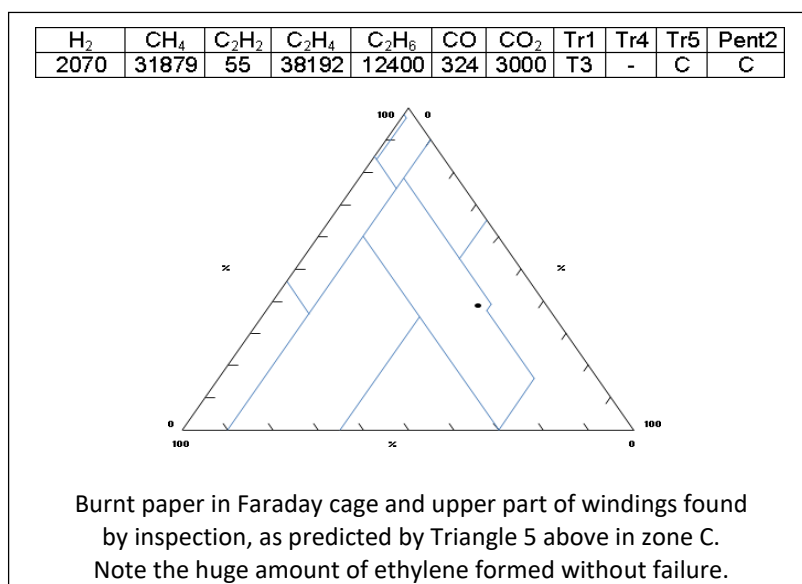
Example 7: Rectifier 120kv 48MVA											
	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	CO	CO ₂	Tr1	Tr4	Tr5	Pent2
2010	13	6	0	69	7			T3	-	T3-H	T3-H
2010	5170	1674	0	380	195	554	5769	T1	S	(T2-H)	S
	(80)	(800)	0	380	195			(T2)	(C)	(C)	(C)
2013	223	2121	0	2883	1708	338	3708	T3	-	C	C

Pentagon 2

Carbonization of paper in windings found by inspection in 2010
and carbonization of leads in 2013

Example 3 of a fault C on windings of a 400 kV voltage transformer⁹ is indicated in Table 3.21.

Table 3.21. Example 3 of a Fault C on Windings⁹



3.10 Carbonization of paper C between turns

Faults C (localized carbonization of paper) between turns can be identified with Triangle 5 (APPENDIX H.2) and Pentagons 2 (APPENDIX H.3).

To determine that a fault C is between turns, acoustic emission or other tests can be used (see APPENDIX C.8). Also carbon monoxide, carbon dioxide and furans (see APPENDIX E.1).

Most important gas formed: ethylene.

Concentration levels of ethylene for localized carbonization of paper between turns are indicated in Table 3.22.

Table 3.22. Concentration Levels of Ethylene for Faults C between Turns

C ₂ H ₄ Conc. Level	Typical	Intermediate 1	Intermediate 2	Pre-failure PF
ppm	2	4	7	25

For definition of gas concentration levels in Table 3.22, see APPENDIX C.4.1 and C.5.

Concentration levels for gases other than ethylene remain the same as in App Table C.7 of APPENDIX C.

At these low levels of C₂H₄, a change in type of stress or fault, e.g., from overheating O to fault C, can be used to decide on trying to locate it in the transformer.

If the fault cannot be confirmed as being between turns, use App Table C.7 also for ethylene.

See APPENDIXES C.4.2, C.5.3 and C.6 on how concentration levels of ethylene in case of faults carbonization of paper C between turns were calculated in Table 3.22.

For recommendations based on gas concentration levels see APPENDIX C.8.

Faults or stresses C with localized carbonisation of paper between turns are probably the most dangerous of faults C, because carbonized paper there is subjected to a high voltage and has lost its electrical insulating properties, resulting in possible arcing D1 and dielectric failure.

The small amount of paper involved in such faults produces a small amount of gas. Only the change from overheating O to fault C will send a warning signal.

Example 1 of localized carbonization of winding turn of reactor¹⁰ is indicated in Table 3.23. Values of CO and CO₂ were not significant, so they are not indicated.

Table 3.23. Example 1 of Fault C in Winding Turn¹⁰

Date	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆		Date	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	
feb13	28	30	0	6	20	●	feb13	28	30	0	6	20	●
feb16	8	17	0	7	17	●	feb16	8	17	0	7	17	●
se17	10	25	0	10	17	●	se17-feb16	2	8	0	3	0	●
dec17	40	29	17	27	17	●	dec17-se17	30	4	17	17	0	●

Pentagon 2

Pentagon 2

Note to Table 3.23: Pentagon 2 on the right corresponds to the Table of values in the upper right. Carbonization of paper occurred in Sept, as predicted by Pentagon 2 in the middle (blue dot), followed by failure in Dec (yellow dot).

A washer in the oil cooling duct (design feature) resulted in the localized overheating and carbonization of paper on a single turn of the middle windings, dielectric failure of the carbonized paper and final arcing.

Fault was initially wrongly attributed to the low DP of surrounding paper (200) on the winding turn.

Example 2 of localized carbonization of paper between turns [B12]¹¹ is indicated in Table 3.24.

Table 3.24. Example 2 of Fault C in Winding Turns¹¹

Date	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	
ma03	34	12	0.9	5	57	●
no09	32	10	0	6.6	25.3	●
ap10	37.2	32.1	0	33.4	33.3	●
delta	5.2	22.1	0	26.8	8	●
ma10	29	9	7	1	17	●
de10	340	102	337	526	102	●

Pentagon 2

Fault was due to inadequate inter-strand insulation, resulting in local carbonization of paper in April 2009, predicted by Pentagon 2 (yellow dot in zone C), followed in Dec by arcing and “hole in winding” fault. Three sister transformers failed similarly.

Example 3 of localized carbonization of paper between turns [B13]¹¹ is indicated in Table 3.25.

Table 3.25. Example 3 of Fault C on Turn¹¹

Date	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	
se91	3	26	0	6	2	●
fe01	6	46	0	9	12	●
fe05	17	59	0	7	16	●
ap09	12	10	0	12	6	●

Pentagon 2

Carbonization of paper C, predicted by Pentagon 2, moved toward a higher temperature fault C (yellow dot) and zone D2 in April 2009, just before failure.

Inspection revealed carbonization of turns in the middle of the upper part of windings, because of inadequate cooling design, followed by shorted turns and arcing failure.

Similar cases have been reported in [B14]-[B16].

In examples 1-3, failure occurred well below typical values of IEC or App Table C.7, and therefore were not due to an IEC fault but to an IEC stress (see APPENDIX A).

3.11 Fault D1 in paper

Faults D1 can be identified with Triangle 1 (APPENDIX H.1) and Pentagons 1, 2 (APPENDIX H.3). Also with the IEC, Rogers and Dornenburg methods.

To determine that a fault D1 is in the paper of windings, acoustic emission or other tests can for instance be used (see APPENDIX C.8). Also carbon monoxide, carbon dioxide and furans, if enough paper is involved (see APPENDIX E.1).

Most important gas formed: acetylene.

Concentration levels of acetylene for faults D1 in paper of windings are indicated in Table 3.26.

Table 3.26. Concentration Levels of Acetylene for Faults D1 in Paper of Windings

C ₂ H ₂ Conc. Level	Typical	Intermediate 1	Intermediate 2	Prefailure PF
ppm	1	3	6	45

For definition of gas concentration levels in Table 3.26, see APPENDICES C.4.1 and C.5. Concentration levels for gases other than acetylene remain the same as in App Table C.7 of APPENDIX C.

If the fault D1 cannot be confirmed as being in the paper of windings, use App Table C.7 for acetylene.

See APPENDIXES C.4.2, C.5.3 and C.6 on how concentration levels of acetylene in case of faults carbonization of paper C between turns were calculated in Table 3.26.
For recommendations based on concentration levels see APPENDIX C.8.

Faults or stresses D1 in paper of windings are more dangerous than faults D1 in oil, because paper there is often subjected to a high voltage and will lose its electrical insulating properties when carbonized by the arcing D1, resulting in dielectric failure. Indeed, ~ 8 cases of faults D1 in paper have been reported to the WG, where failure occurred when acetylene reached 120 or 45 ppm or less.

Example of sparking partial discharges D1 in paper of a 230 kV bushing¹ is indicated in Table 3.27.

Table 3.27. Example of Sparking Partial Discharges D1 in Paper

	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	Tr1	Tr4	Tr5	Pent2	
01 April	576	150	11	116	19	DT	S	T3	S/D1	●
15 April	433	115	9	92	15	DT	S	T3	S/D1	●
15 May	966	226	21	179	32	DT	S	T3	S/D1	●
12 June	1212	225	21	171	30	DT	S	T3	S/D1	●
Delta 15 Apr/15 May	533	111	12	87	17	DT	S	T3	S/D1	●
Delta 15 May/12 June	246	0	0	0	0	-	S/PD	-	PD	●

Pentagon 1

Note: as seen in photos above, this case had multiple faults: partial discharges of the sparking type D1 in paper (carbonized punctures), T3-H fault on metal part in oil, and stray gassing S of surrounding oil (see APPENDIXES H.6 and H.7).

3.12 Fault D2 in windings

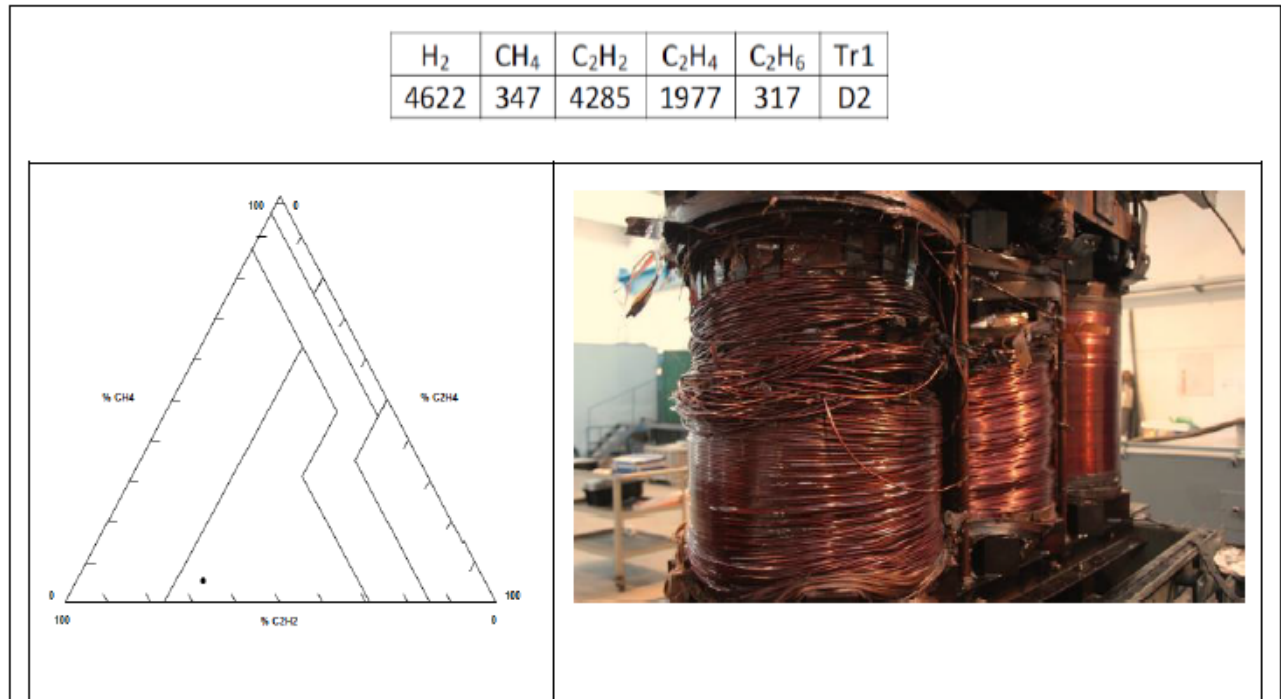
Faults D2 can be identified with Triangle 1 (APPENDIX H.1) and Pentagons 1, 2 (APPENDIX H.3). Also with the IEC, Rogers and Dornenburg methods.

The most important gas formed is acetylene.

Faults D2 often occur suddenly, producing large amounts of gases, Buchholz alarms and automatic tripping of the transformers. Confirmed faults D2 are the most dangerous of all faults in transformers. However, small amounts of gas of the D2-type in transformers are often rather due to less dangerous mixtures of stresses D1 and T3 (see APPENDIXES H.6 and H.7).

Example of a fault D2 with axial collapse of windings⁷ found by inspection is indicated in Table 3.28.

Table 3.28. Example of Fault D2 Identified with Triangle 1⁷ and Axial Collapse of Windings found by Inspection



4. How to detect faults in on-load tap- changers with DGA

4.1 Duval triangles 2 for DGA in on-load tap-changers

The use of DGA in on-load tap-changers of the compartment and in-tank types has been described in [B17], [B18], CIGRE TB 443 [B1] and IEEE C57.139 [B19], and corresponding Duval Triangles 2 have been developed for that purpose.

In CIGRE TB 443, a classification scheme (AXC, AXS, VXC, VXS, ARC, ARS, VRC, VRS) has been introduced to distinguish between different on-load tap-changer architectures [B2]. For each architecture, a typical gassing pattern for fault-free operation can be defined. The diversity of gas patterns observed in service necessitates the definition of additional “Normal zones” N1, N3, N4, N5, as shown in Triangle 2a, see Figure 4.1.

Compartment type on-load tap-changers (class AXC, AXS, VXC, VXS) show their normal gas pattern in zone N of Triangle 2. This is also true for some in-tank on-load tap-changers types of the ARC and ARS class. But these types can also produce gas patterns in the N1, N4 or N5 zone without showing a malfunction. This depends on the individual operating conditions and time in service [B17], [B17].

In-tank vacuum type on-load tap chander models (class VRC, VRS) usually show their normal gassing pattern in zone N3. It has to be noted that the absolute values of hydrocarbon gases observed in these models are very low and so often don't allow a distinct assignment to zone N3.

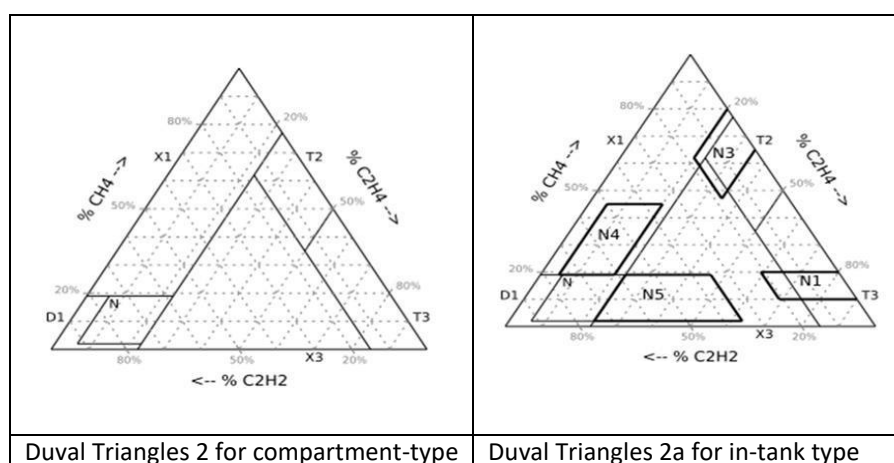


Figure 4.1. Duval Triangles 2 and 2a for On-Load Tap-Changers of the Compartment and In-Tank Types, respectively [B19]

Models of compartment-types on-load tap-changers with normal gas formation occurring in zone N of Triangle 2 are for example those of ABB (UC, UD, UV, UZ), GE (LR), Westinghouse (UN, UR, UT), Siemens (TLH), Ferranti (RT), Mc-Graw Edison (BLS), Cooper (Regulators) etc [B17]. Faulty operation occurs mostly in zones T3/T2 (coking of contacts). Faults T3/T2 in progress and abnormal arcing D2 occur in zone X3, and abnormal arcing D1 in zone X1 or N.

Faulty operation can be suspected when gas formation occurs outside of the normal zone of a given on-load tap-changer model. To confirm faulty operation, however, the amounts (in ppm) of the main gases formed per operation of the OLTC must be significantly higher (e.g., >50%) than those observed during normal operation [B2].

Thermal stresses on transition resistors (in on-load tap-changer models of class ARC, ARS, VRC, VRS) have in some cases been detected by very high levels of carbon monoxide and carbon dioxide, because of accelerated oil oxidation¹¹. Especially for vacuum type models with very low gas generation, carbon monoxide and carbon dioxide can act as early indicator of thermal oil ageing, caused by the transition resistor heating [B20].

4.2 Normal and faulty operation of in-tank types

Table 4.1 shows a compilation of normal and faulty operation observed with different types of in-tank on-load tap-changer models of Maschinenfabrick Reinhausen GmbH in different countries, presented by WG members.

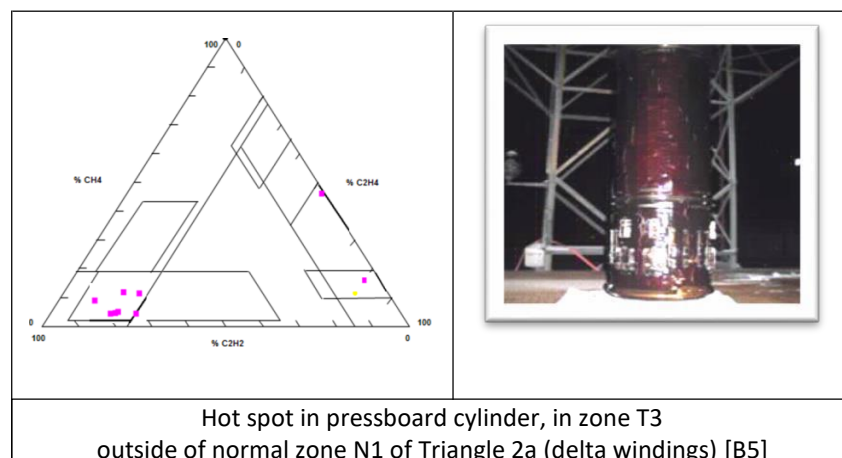
**Table 4.1. Normal and Faulty Operation of In-Tank Type
On-Load Tap-Changers in Various Countries**

Type	Normal operation in zone:	Faulty operation in zone:
M	N, N1 or N5	X3 (6 arcing, 2 thermal) T3 (6 thermal,)
R	N then N1, N5 or N4 often in N1	X3 1 coking) T2 (2 thermal)
D	N1 or N4	T2 (1 thermal)
G	N or N5	
T	N then N1	
V	N	
VR	N3	
VV	N or N3	T3 (1 thermal)

Examples of faulty operation of some units of the in-tank type are presented in Table 4.2 to Table 4.5:

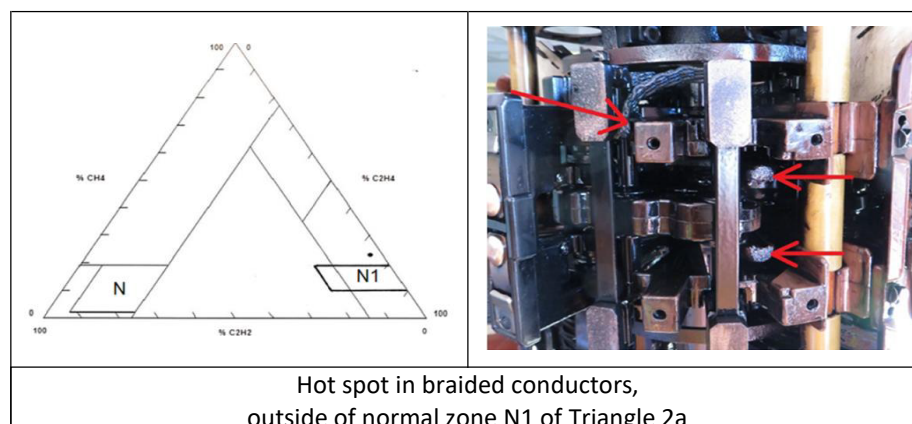
Example 1 of faulty on-load tap-changer of class ARS type D is indicated in Table 4.2:

Table 4.2. Example 1 of Faulty On-load Tap-Changer of Class ARS



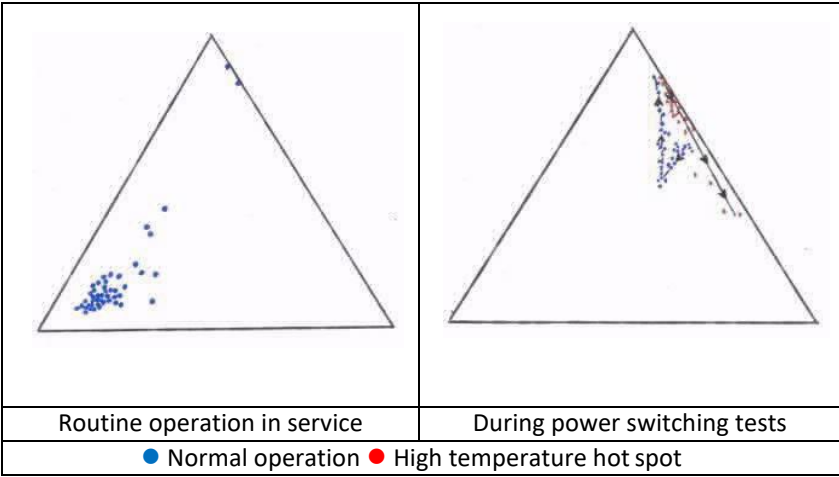
Example 2 of faulty on-load tap-changer of class ARS type M is indicated in Table 4.3.

Table 4.3. Example 2 of Faulty On-Load Tap-Changer of Class ARS



Example 3 of faulty on-load tap-changer of class VRC is indicated in Table 4.4.

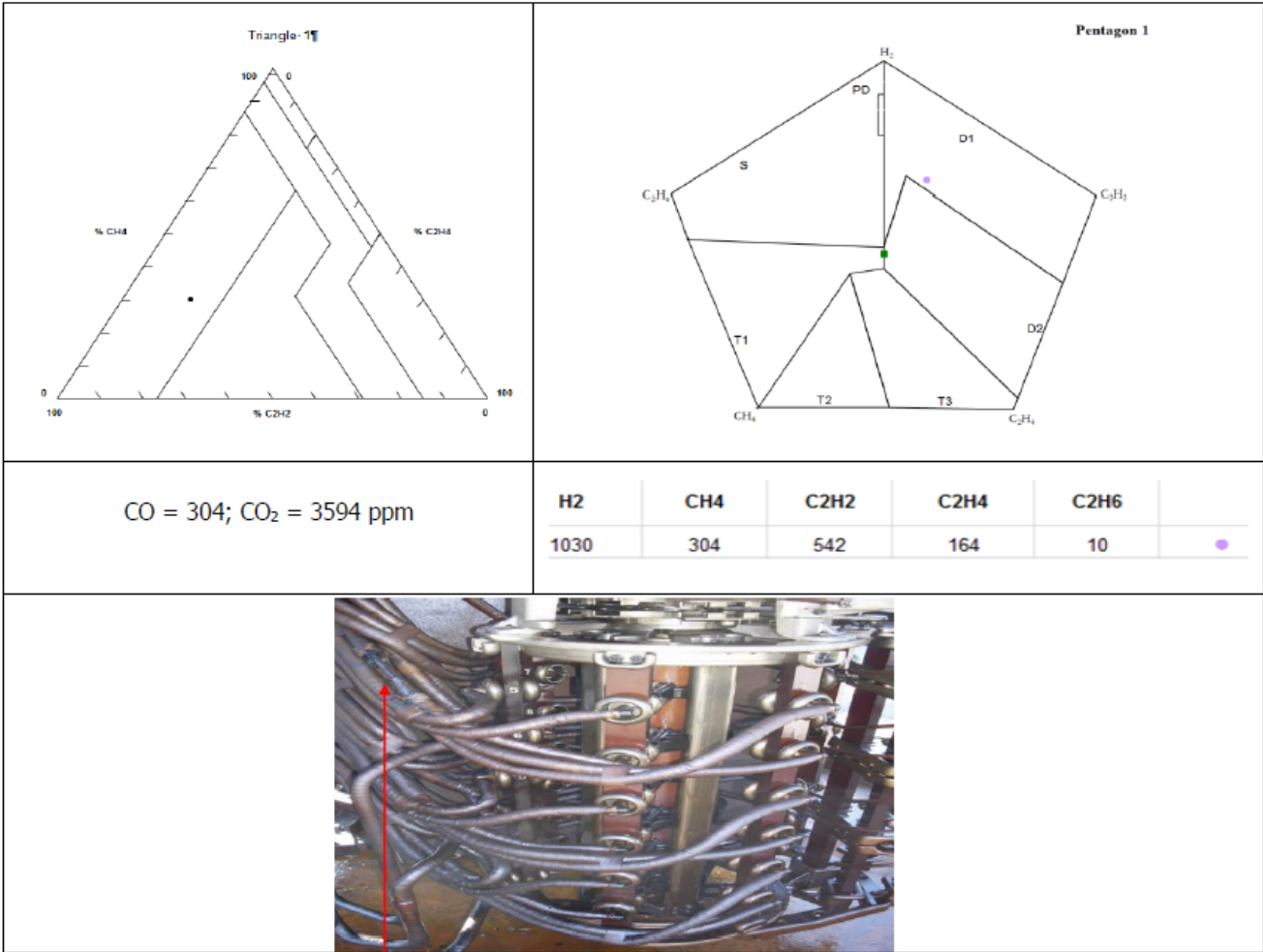
Table 4.4. Example 3 of On-Load Tap-Changer of Class VRC



Note to Table 4.4: Such gas formation in this on-load tap-changer [B17], was due to insufficient grounding of one shielding ring, and has no longer been observed in on-load tap-changers manufactured later than 2012.

Example 4 of faulty on-load tap-changer producing arcing D1 in the main tank of the transformer, in conductor to windings and gas alarm, detected by Triangle 1 and Pentagon 1, is indicated in Table 4.5.

Table 4.5. Example 4 of Faulty On-Load Tap-Changer Producing Arcing D1 in Transformer Main Tank

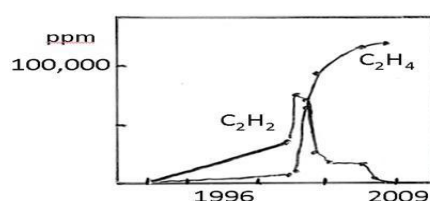


Another method using propylene, ethylene and methane (the PEM method, see APPENDIX I.3) has also been proposed in Thailand and Japan for identifying the normal and faulty operation of on-load tap-changers [B37], allowing evaluation of the temperature of oil during a gas-producing fault. This temperature is based on theoretical thermodynamic considerations, not on actual measurements of temperature in on-load tap-changers. It is limited to DGA laboratories and countries reporting C_3 hydrocarbons.

4.3 Factors affecting normal gas formation of in-tank types

The main factors affecting normal gas formation of in-tank types, for a given model of on-load tap-changer, are time in operation and power level [B21]. “Power level” includes the model scale/size, the actual load and operation frequency (number of operations per time interval).

The evolution of gas formation over time in operation often observed on ARS class on-load tap-changer models is shown for example in Figure 4.2 [B31]. During the first few months or years of operation, normal gas formation occurs in zone N with acetylene as the major gas. This has to be expected due to the normal arc-breaking activity of the switching contacts. With increasing time and number of operations, the gas pattern moves towards zone N1, with mostly ethylene formation and without any sign of a fault (verified by inspection).



**Figure 4.2. Evolution of Normal Gas Formation
of ARS class on-load tap-changer with Time in Operation [B21]**

Tests have shown that the arcing time and arcing energy don't show significant changes between fresh and carbonized oil¹².

Several possible reasons for the evolution of normal gas formation in in-tank types from normal zone N to normal zones N1 have been investigated by the WG:

- The normal arc-breaking activity causes severe oil deterioration and generates huge amounts of carbon particles (soot). A significant portion remains in suspension in the oil and causes changes to the electrical and thermal conductivity. It is assumed that these changes cause different heat dissipation or gradient of the switching arc, favoring the formation of ethylene instead of acetylene.
- It is also hypothesized that the shape of the switching arcs in carbonized oil is more “distributed” and so causes lower maximum temperatures (below 1000°C), which favors the formation of ethylene as above.

These assumptions are supported by observations that, after cleaning the oil compartment and replacing the old oil by a new one, mostly acetylene is again formed. However, it is being replaced by ethylene more rapidly than when the on-load tap-changer was first put in service.

This is also partially supported by observations in Indonesia¹ on ARS class on-load tap-changers where the evolution of gases moved from zone N to N1 when the number of operation per day was changed from 10 to 25.

Thermodynamic considerations [B22] studied in Brazil, and illustrated in Figure 4.3, reveal that higher oil temperatures (> 1200°C) form gases in the left part of Duval Triangle 1 and Triangle 2 (in the N zone), while lower oil temperatures (down to 600°C) form gases rather in the right part (zones N1- N3-T3). Distributed switching arcs could for example heat-up larger volumes of oil around the arc and so decrease the average oil temperature. This is also what is observed in transformers in case of high-energy discharges of lower apparent temperatures in zone D2, compared to low-energy discharges of higher apparent temperatures in zone D1¹ (see Chapter 2.3, Note 1).

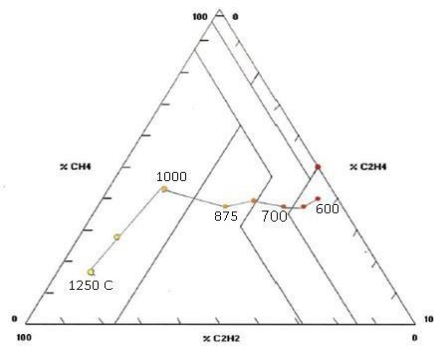


Figure 4.3. Thermodynamic Considerations vs Gas Formation in Duval Triangle 1 [B22]

A second possible reason for an increased ethylene content could be desynchronized contacts, which have been observed in Canada¹ and in the US¹³ to create hot spots in on-load tap-changers. Desynchronized, broken or worn diverter or selector switch contacts prolong the arcing time and so the length of the switching arc (keeping the total arcing energy unchanged), which involves a greater oil volume but that is heated to a lower maximum temperature.

Detailed information about the respective shape of the switching arcs, arc duration, contact resistance and changes in thermal and electrical conductivity could help verify these observations and assumptions. Appropriate instruments (for acoustic tests and recording of RF signals) have been identified that could provide such information.

5. Conclusions

Ten different types and sub-types of faults in electrical equipment are described in this TB, detectable with Triangles 1, 4, 5 and Pentagons 1 and 2.

Among these faults, stray gassing of oil S, partial discharges of the corona-type PD, overheating O < 250°C, high temperature faults in oil only T3-H and arcing faults D1 in oil only, tend to produce larger amounts of gases in transformers without affecting their normal operation (higher typical and pre-failure values), and are of relatively lesser concern in transformers.

Carbonization of paper C and arcing D1/D2 in paper are potentially more dangerous faults, producing lower amounts of gases before becoming a threat, particularly faults C between turns. Faults C in leads are a bit less of a risk.

Power transformers with low oxygen to nitrogen ratios in oil tend produce more faults of the low-temperature-type (S, O, T1), but these faults are of lesser concern in transformers.

Wind-farm transformers and bushings tend to produce more hydrogen and faults S and corona PD than power transformers, but these also are of lesser concern.

Stray gassing of oil produces mostly hydrogen in mineral and silicone oils, ethane in ester oils. Low-temperature faults in paper have not been observed to significantly affect the failure rate of transformers.

Multiple faults can be detected by combinations of the Triangles and Pentagons.

Reasons for the evolution of normal gas formation in on-load tap-changers of the in-tank-types with years in service are still under investigation.

APPENDIX A. Definitions, abbreviations and symbols

A.1. Terms and definitions

- **fault** [IEV 604-02-01]: “an unplanned occurrence or defect in an item which may result in one or more failures of the item itself or of other associated equipment”.

Note: gas formations below typical values are not considered in IEC 60599 [B3] as due to a “fault”. However, it has been observed by WG D1/A2,47 that some faults and failures can indeed occur well below typical values, so the IEC definition will need to be amended.

- **typical values** [IEC 60599]: “ values of gas concentration or rates which have no symptom of fault or failure and which are overpassed by only arbitrary percentage of higher value of concentrations or rates” .

Note: typical values for power and instrument transformers at the IEC and CIGRE are based on 90% percentile values, and for bushings 95%.

- **stress** [IEC 60599 , Section 5]:” any gas formation in service, be i t minimal, results from a stress of some kind [type], even i f i t i s a very mild one , l i ke normal temperature ageing . However, as long as gas formation is below typical values, i t should not be considered as an indication of a “ fault”. “
- Note: in some cases, failures occur well below IEC typical values and therefore are not due to IEC faults but to IEC stresses (see for instance section 3 . 10).

APPENDIX B. Links and references

- [B1] CIGRE Technical Brochure 296 (2006) Recent Developments in DGA Interpretation, CIGRE WG D1.01/ A2.11, www.e-cigre.org
- [B2] CIGRE Technical Brochure 443 (2010) DGA in Non-Mineral Oils and Load Tap-Changers and Improved DGA Diagnosis Criteria, CIGRE WG D1/A2-47, www.e-cigre.org
- [B3] IEC Publication 60599, Mineral Oil-Impregnated Electrical Equipment in Service – Guide to the Interpretation of Dissolved and Free Gases Analysis
- [B4] Draft Technical Brochure (to be published in 2019) DGA Monitoring Systems, CIGRE WG D1/A2.47, www.e-cigre.org
- [B5] Use of Duval Pentagons and Triangles for the Interpretation of DGA in Electrical equipment, TechCon North America Conference, Albuquerque, 2016, M.Duval
- [B6] Dissolved Gas Analysis: a Powerful Maintenance Tool for Transformers, TechCon North America Conference, San Antonio, 2004, M. Duval
- [B7] Improving the Reliability of Transformer Gas-in-Oil Analysis, IEEE EI Magazine, Vol.21, No 4, pp.21-27, 2005, M.Duval and J.Dukarm
- [B8] Identifying and Analyzing Quick Developing Faults with DGA, SIGAT Conference, Cartagena, Colombia, 2014, M.Duval
- [B9] IEC Publication 60628-A, Gassing of insulating liquids under electrical stress and ionization [B10] ASTM Standard Test Method D 2300, Gassing of Electrical Insulating Liquids Under Electrical Stress and Ionization
- [B11] Significance and Detection of Very Low Degree of Polymerization of Paper in Transformers, IEEE EI Magazine, Vol.33, No 1, pp.31-38, 2017, M.Duval, A.dePablo, I. Hohlein-Atanasova and M.Grisaru
- [B12] Tranaformer Asset Management: achievements and prospects for further improvements, CIGRE SC A2 Colloquium, Shanghai, 2015, H.Ding, S.Ryder, P.Jarman and G.Stevens [B13] Why Transformers Fail, Euro TechCon, 2009, H.Ding, R.Heywood, J.Lapworth and S.Ryder
- [B14] Diagnosing Difficult Transformer Problems Using On-Line Condition Monitoring, CIGRE SC A2, Paper A2-108, 2016, H.Ding, R. Heywood, P.Jarman, S. Ryder and S.White
- [B15] Testing Transformers in the Field, Doble-Life of a Transformer Seminar, 2013, R.Zaleski, J.Lapworth and S.Ryder
- [B16] Learning from Power Transformer Forensic Investigation and Failure Analysis, CIGRE SC A2, Paper A2-109, Paris, 2014, E. Cross, H.Ding, R.Heywood, R.Hooton, S.Horsley P.Jarman, J.Lapworth and S.Ryder
- [B17] Application of Duval Triangles 2 to DGA Analysis in Load Tap-Changers, EuroTechCon Conference, 2012, M.Duval
- [B18] Use of Duval Triangles and Pentagons for DGA in Transformers, LTCs and Non-Mineral Oils, AsiaTechCon Conference, Sydney, Australia, 2015, M.Duval
- [B19] IEEE Transformer Committee (2015), IEEE Guide for Dissolved Gas Analysis in Transformer Load Tap-Changers, IEEE C57.139
- [B20] Carbon Oxides in the Interpretation of Dissolved Gas Analysis in Transformers and Tap- Changers, IEEE Elec.Insul.Magazine, Vol. 26, No.6, pp. 22-26, 2010, I.Hoehlein-Atanasova and R.Frotscher
- [B21] Update on the Interpretation of DGA in HV Equipment, EPRI-TSUG Conference, Philadelphia, 2015, M. Duval
- [B22] Ongoing Activities at IEEE, IEC and CIGRE on DGA, EPRI-TSUG Conference, St-Louis, 2013,

M.Duval

- [B23] Calculation of DGA Limit Values and Sampling Intervals in Transformers in Service, IEEE EI Magazine, Vol.24, No 5, pp.7-13, Oct 2008, M.Duval
- [B24] CIGRE Technical Brochure 409 (2010), Report on Gas Monitors for Oil-Filled Electrical Equipment, CIGRE WG D1.01/TF 15
- [B25] Utilizing Piecemeal Linear Approximation and Harmonic Regression to Analyze Power transformer Insulating Oil On-Line Gas Samples, TechCon North America, 2010, D.Lamontagne
- [B26] Dynamic Behaviour of Fault Gases and Online Gas Sensors, Paper A2-116, CIGRE Meeting Paris, 2016, S.Tenbohlen, I.Hoelein, M.Lukas, A.Muller, K.Schroder and U.Sundermann
- [B27] Reduction of failure risk in power transformers through the detection and location of incipient faults using acoustic emission, TechCon North America Conference, StPetersburgh, Florida, 2003, A.Nunez, R.K. Miller and B. Ward
- [B28] Health Indices and Life Assessment Methodology, Transformers Magazine, Vol.3, No.3, 2018, M.Grisaru
- [B29] M/DBT, New Alternative Dielectric Liquids for Transformers, Paper D1-107, CIGRE Symposium, 2012, J.Walker, A.Valot, Z.D.Wang, X.Yi and Q.liu
- [B30] A novel Approach for Bushing Fault Diagnosis; Power Grid India Experience, CIGRE SC A2 paper A2-208, Paris, 2018, S.Harichandranray, J.Das, R.Tyagi and P.N. Dixit
- [B31] Experimental Evaluation of Status of 400 kV Shunt Reactor Bushings in the Swedish National Grid, CIGRE SC A2 Paper A2-203, Paris 2018, L. Jonsson, L.Melzer, N. Schonborg and G-O. Persson
- [B32] Transformer Oil fault Gases Under Thermal Stress at 160°C – Part II, Transformers Magazine, Vol. 5, No. 3, 2018, B. Christian and A. Glaser
- [B33] Mineral Insulating Oil Stray Gassing and its Effect on DGA Laboratory Study, TechCon North America Conference, Phoenix, 2014, J.Weemann, M.Stern, B.Pahalavanpour and J.Nunes
- [B34] Stray Gassing of Transformer Insulation Oils - Impact of Materials, Oxygen Content, Additives, Incubation Time and Temperature, and its Relationship to Oxidation Stability, IEEE-EI Magazine, Vol.33, N0.6, pp.27-32Nov/Dec 2017, S.Eeckhoudt, S.Autru and L.Lerat
- [B35] Interpretation of Gas-In-Oil Analysis Using New IEC Publication 60599 and IEC TC 10 Databases, IEEE-EI Magazine, Vol. 17, N0 2, pp 31-41, 2001, M.Duval and A.de Pablo
- [B36] Improvement of Duval Triangle 1, CIRED Conference, Glasgow, 2017, S.Spremic
- [B37] New Dissolved Gas Analysis (DGA) Diagnostic Method for Load Tap-Changers (LTC), CIGRE A2 & D1 Joint Colloquium, Kyoto, 2011, P.Kuansatit and A.Tong-in
- [B38] Tap-Changer DGA – Uncovering an Enigma, Part 1, Transformers Magazine, Vol. 4, 3, 2017, R. Frotscher
- [B39] Performance of Oil and paper in Transformers based on IEC 61620 and Dielectric Response Techniques, IEEE-EI Magazine, Vol. 26, No 3, pp. 16-23, 2010, S. Bhumiwat, P. Wickramasuriya and P. Kuansatit
- [B40] The New Duval pentagons Available for DGA Diagnosis in Transformers Filled with Mineral and Ester Oils, Elec.Insul.Conf., Baltimore, June 2017, M.Duval and L.Lamarre

Trademarked products

Jarylec® GA M/DBT mono and dibenzyltoluene additives in oil (see APPENDIX D.1). Irgamet 30 and 39 passivator additives in oil (see APPENDIX G.3).

Acknowledgements

¹ M.Duval, ² S.Bhumiwat, ³ E. Alzieu, ⁴ W.Johnson, ⁵ A.M.Dale, ⁶ A.Constant, ⁷ O.Amirouche, ⁸ A. Nunez, ⁹ M.A. Martins, ¹⁰ A. Fieldsend-Roxborough, ¹¹ S.Ryder, ¹² R.Frotscher, ¹³ P.Boman, ¹⁴ B.Nemeth, ¹⁵ F.Scattigio, ¹⁶ C.Beauchemin, ¹⁷ S. Leivo, ¹⁸ J.Rasco, ¹⁹ T.Buchacz, ²⁰ I. Atanasova-Hoehlein, ²¹ A.de Pablo, ²² A. Hadzi-Skerlev, ²³ S. Dorieux, ²⁴ J. Wang, ²⁵ P.deBijl, ²⁶ M. Grisaru, ²⁷ J. Lukic, ²⁸ S.Spremic, ²⁹ D.Rodda, ³⁰ R.Baldyga, ³¹ T. Dalton, ³² T. Krieg, ³³ E. Hocma, ³⁴ T. Gurley, ³⁵ M. Banovic.

APPENDIX C. Calculations on the DGA database of WG D1/A2.47

C.1. DGA database of the WG

C.1.1. Individual DGA databases

To calculate the influence of type and location of fault or stress on gas levels, the WG has created a large database of DGA results including several individual databases provided by its members.

To ensure the confidentiality of data, the name of the transformer owner and transformer manufacturer, as well as any internal identification number, were removed from the merged database. Furthermore, access to it was given to only one member ¹⁴ of the WG, who was in charge of doing calculations on it, and subject to a strict confidentiality agreement of not circulating the merged and individual databases to anyone, and of destroying it after the investigations by the WG on it had been completed.

The purpose of the merged database was to evaluate the influence of type and location of faults on percentile values of concentrations in ppm, not the influence of other factors such as voltage, MVA, age or operating conditions of transformers. The basic information required in individual databases was therefore limited to raw DGA results in ppm, date of oil sampling for analysis, an anonymous number identifying each transformer and, when available, whether there was an on-load tap-changer communicating or leaking into the main tank, and whether the oil was of the mineral or non-mineral type. All other information in the individual databases (voltage, etc) was deleted either by WG members providing them, or by the WG itself.

App Table C.1 contains a list of the individual DGA databases provided by WG members. These individual databases were cleaned ¹⁴ of inappropriate data such as those measured in non-mineral oil or obtained from on-load tap-changers, duplicated data, values of 0 ppm were replaced by 1 ppm, etc. They were then merged into the database of the WG, after another round of clean-up.

C.1.2. Representativity of the DGA database of the WG

It has been a common practice of IEC, IEEE and CIGRE, during the past 50 years, to use big DGA databases to calculate 90% percentile values (called typical values by IEC/CIGRE and condition 1 values by IEEE). All these databases, including those used by IEC to calculate typical values in 60599 [B3], contained DGA results coming by various sources (utilities, industries, laboratories, etc), and from all types of power transformers of different voltages, power levels, ages, manufacturers, etc.

The 90% percentile typical values of gas concentrations calculated on the DGA database of the WG ("All results") are within or very close to the ranges of values in IEC 60599 [B3], as can be seen in App Table C.2, confirming that the database of the WG, despite containing DGA results from various sources and types of power transformers, and not only from utilities, is as representative of gas formation in transformers as the databases used for IEC 60599.

It is sometimes suggested^{10,15} that calculations using all DGA results in a database are not representative of the transformer population, because some transformers in service for a long time or with gassing problems have been sampled for DGA more often than the others. App Table C.2, however, shows that results are not very different when only the latest DGA results for each transformer are used, and are also within or close to the range of values of 60599. "Only latest results" in App Table C.2 means the most recent DGA results reported on each transformer.

This indicates that calculations made with all DGA results in the database, as was done in this TB, are as representative as those using only the latest DGA results.

One major advantage of using all DGA results rather than only the latest ones is that the number of results in the sub-databases derived from them is high enough to allow reliable calculations of typical values to be done.

Note: The database of SDMyers contains some wind farm transformers with a higher typical value of hydrogen. However, it has been verified that the number of wind farm transformers in the database of SDMyers is too small to affect the calculations of its typical values of hydrogen. –

App Table C.1. Individual DGA Databases Used by WG47

Individual Databases of:	Contact	Number of DGA Results
Duke Energy (US)	M.Banovic/ T.Rhodes	83,639
SDMyers (US)	T.Gurley/ L.Parthemore	54,057
Hydro Quebec (CA)	C.Rajotte	50,136
Morgan Schaffer	S.Brauer	40,000
EGAT (TH)	P.Kuansatit	37,689
Endesa (ES)	S.Santaella/ A.de Pablo	28,000
FKH (CH)	M.Koch	27,300
Oksman Seraphin (FR)	C.Epain-Molle	14,072
N.Tesla (RS)	J.Lukic	12,726
KEA (NZ)	S.Bhumawat	12,345
Weidmann	N.Perjanik	9,645
Terna (IT)	F.Scatiggio	7,085
Elma Servizi (IT)	O.Amirouche	4,891
Nucor	Wilson	1,665
Iberdrola (ES)	P.de la Fuente	1,697
Adinet (UR)	M.Banovic	718
Elektrovoyvodina (RS)	S.Spremic	154
Trinidad	M.Banovic	59
TOTAL		335,828

App Table C.2. Typical Values of Gas Concentrations in the Database of the WG and in IEC 60599

Database of WG47 Typical values in ppm	Number of DGA results	H ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	CO	CO ₂
All results	337,805	118	85	111	56	5	700	6300
Only latest results	85,059	81	55	54	48	2	730	6660
IEC 60599 (No OLTC)	N/A	50-150	30-130	20-90	60-280	2-20	400-600	3800-14,000

C.2. Identification by DGA of faults or stresses in the DGA database of the WG

C.2.1. Separation into sub-databases of faults or stresses

For identifying the causes of gas formation (faults or stresses) according to DGA in the merged database of the WG, the merged database was first separated ^{1,14} into 6 sub-databases corresponding to the 6 basic causes of gas formation defined in IEC 60599 and identified with Triangle 1 (D2, D1, T3, T2, T1, PD).

Subdatabases T2, T1 and PD were further separated into 4 additional sub-databases corresponding to the 4 additional or improved sub-causes of gas formation as defined above and identified with Triangle 4 (S, O, PD, C).

Subdatabases T3 and T2 were also further separated into 2 additional sub-databases corresponding to the 2 additional sub-causes of gas formation as defined above and identified by Triangle 5 (T3-H, C).

C.2.2. Possible misidentifications of faults or stresses

The problem of possible misidentification of faults or stresses at low gas concentrations has been investigated.

According to IEC 60599, the accuracy on DGA results >10 ppm is better than $\pm 15\%$, except for a few “bad” outlier laboratories that cannot be identified in any database. The large majority of results >10 ppm, however, can be assumed to meet the accuracy requirements of IEC. Such accuracy allows a reliable identification of faults or stresses when using identification methods such as Triangles 1, 4, 5 and others, as demonstrated in [B7].

Accuracy on DGA results > 5 ppm is around $\pm 30\%$, allowing an identification of faults or stresses in the database which is a little bit more uncertain but still acceptable [B7]. Below 5 ppm, the uncertainty on identifications is much higher and misidentifications are likely.

The uncertainty on the identification of faults or stresses T3 with Triangle 1 is low, even at low ppm values. For instance, in order to have DGA results appearing in zone T3 of Triangle 1, the minimum amount of ethylene necessary is 5 ppm (e.g., with methane and acetylene at 1 ppm and 1 ppm, respectively), a gas concentration still allowing reliable identifications (see above). This was confirmed by typical values of ethylene calculated on the whole database after deleting ethylene values < 5 , 4, 3 or 2 ppm, which remained the same, meaning that they were not affected by values < 5 ppm. So there was no major misidentification of stresses T3 in the database at low ppm values.

The uncertainty on the identification of faults or stresses T2 and T1 in Triangle 1 at low ppm values is even lower, with the minimum amount of ethylene and methane needed to appear in zones T2/T1 being more than 10 ppm, where identification of faults or stresses are quite reliable.

The identification of faults or stresses D1 is less reliable, the minimum amount of acetylene necessary to appear in zone D1 of Triangle 1 being 5 to 3 ppm (for instance acetylene/ethylene/methane = 3/1/1 ppm). For typical values of acetylene calculated on the whole database not to be affected by low values of acetylene, values < 2 ppm had to be deleted in the sub-database D1. Since 90% of values of acetylene in many individual databases are < 2 or 1 ppm, this eliminates a large amount of results which would otherwise be misinterpreted as causes D1.

The uncertainty on the identification of faults or stresses in zones D2 and DT of Triangle 1 at low ppm values of the three gases is much higher. Typical examples of such cases are for example (2, 2 and 2 ppm of the three gases). This corresponds to DGA results with virtually no gas formation, occurring in a very large number of cases, almost half of them in big databases. Also possibly resulting in misidentifications in these zones are mixtures of causes T3+ D1 or T2+D1. So, no attempts have been made to do calculations on DGA results in zones D2 and DT of Triangle 1.

Concerning faults or stresses D1 again, it has also been observed that some individual databases of App Table C.1 have high typical values of acetylene (above 2 or 5 ppm). In IEC 60599, such high typical values have been associated with communicating on-load tap-changers rather than to real faults or stresses D1. Since no mention was available in individual databases on whether or not some of them had a communicating on-load tap-changer, and in order to avoid misidentifying apparent faults or stresses D1 actually due to contaminations by on-load tap-changers or other reasons, only individual databases with low typical values of acetylene were kept for the identification of faults or stresses D1. Deleting values of acetylene < 2 ppm in the sub-database D1 then totally eliminated the problem of misidentification of faults or stresses D1 in the sub-database.

Note: it has been observed that large DGA databases used in North America usually have low typical values of acetylene (e.g., < 2 or 1 ppm). Some of them, however, have significantly higher values,

either because they do not degas the oil after having repaired an arcing fault, or because some of their transformers have unidentified communicating on-load tap-changers.

C.2.3. Occurrence of faults or stresses in transformers

The occurrence of faults or stresses according to DGA in the database of the WG has been calculated as the number of DGA results in each sub-database, divided by the total number of DGA results in the whole database, expressed in % of DGA results in the database, and has been indicated in App Table C.3. As mentioned above, for more than 50% of DGA results in the database, no reliable cause of gas formation can be calculated because gas levels are too low, particularly in zones D2 and DT of Triangle 1, explaining why the total % of faults or stresses in App Table C.3 is less than 100%.

C.2.4. Typical gas concentrations vs types of faults or stresses in transformers

Typical 90% percentile gas concentration values have been calculated for each sub-database of faults or stresses of the WG.

Only individual databases of App Table C.1 with 90% percentile values for acetylene < 1 ppm were kept in the sub-database D1 for calculating the typical value of acetylene. Acetylene values < 2 ppm were also deleted in the resulting sub-database D1, for the reasons explained in C.2.2.

Values for partial discharges of the corona-type PD have been calculated with Triangle 4, not with Triangle 1, which may contain some faults or stresses S in addition to partial discharges of the corona-type PD in its PD zone.

C.3. DGA occurrence and severity of faults and stresses in the dga database of the WG

C.3.1. Occurrence of types of faults or stresses identified by DGA

The DGA database of the WG has been used to calculate the occurrence of faults (above typical values) or stresses (below typical values) in transformers. Results in % are indicated in App Table C.3.

In order to do that, this DGA database has been separated into sub-databases corresponding to each of the 10 sub-types of faults or stresses of Table 2.1 in Chapter 2. For details on the calculations made, see C.2.3.

App Table C.3. Occurrence of Types of Faults or Stresses identified by DGA in the DGA Database of the WG^{1, 14}

Fault or Stress Identified by DGA in the Database, Using:		Type of Fault or Stress	Occurrence of Type of Fault or Stress, in %
Subdatabases:	Triangles:		
T1+T2+T3+D1+D2+PD	1	T3	22
		T2	8
		T1	14
D1	1	D1	1
T2+T3	5	T3-H	21
		C	6
T1+T2+PD	4	S	8
		O	7
		Corona PD	0.04

Note: App Table C.3 indicates the % of faults (above typical values) and stresses (below typical values) identified by DGA in the database of the WG. The % of types of faults only (above typical values) is much smaller than in App Table C.3 (about 10 times lower). For instance there are 2.2 % of faults T3 in transformers (above typical values).

It can be seen in App Table C.3 that faults or stresses occurring in transformers are mostly of the thermal type T3, T2 and T1. These thermal faults or stresses are occurring mostly in oil only (T3-H, S) or at low temperatures (O), and therefore are not a major concern for transformers.

Faults or stresses potentially dangerous for transformers (C, D1) are observed only in a minority of cases. This probably explains why there are so few failures of transformers in service (in only about 0.3% of transformers annually, as indicated in [B4]).

The influence of the oxygen/nitrogen ratio on the occurrence of faults or stresses identified by DGA is indicated in App Table F.2 of APPENDIX F.

C.3.2. Severity of types of faults or stresses

The severity of types of faults (above typical values) or stresses (below typical values) in transformers depends on their location (in paper or in oil), as summarized qualitatively in App Table C.4.

- the less severe types of faults or stresses are those of lower temperature (S, O, T1, PD) and/or in oil only (S, T3-H, D1 in oil). They are of lower concern for the equipment (the probability of having a failure with such faults is low or very low).
- the most severe types of faults or stresses are those of higher temperature or energy (D2) and/or occurring in paper (C, D1 in paper). They are potentially the most dangerous for the equipment and require more attention and monitoring (the probability of having a failure is higher).

For instance, faults or stresses T2 in paper insulation at $T > 300\text{ }^{\circ}\text{C}$ will produce carbonization C of paper, resulting in the loss of its electrical insulation properties, possible electrical breakdown and failure (see Chapters 3.8 to 3.10).

As indicated in APPENDIXES H.2 and H.3, Triangles 4, 5 and Pentagon 2 can detect a possibility of carbonization of paper C, but not with a 100% certainty [B5]. It is therefore recommended to verify that possibility with carbon monoxide, carbon dioxide and furans.

App Table C.4. Severity of Types of Faults or Stresses[B22]

FAULT :	IN PAPER		IN OIL	
TYPE :	Main products formed	Severity	Main products formed	Severity
D2	C, C ₂ H ₂	Very high	C ₂ H ₂ , C	Very high
D1				Moderate
T3	C, C ₂ H ₄		C ₂ H ₄ , C	
T2	C, CH ₄	High	CH ₄	Low
T1, O	C ₂ H ₆ , CO	Moderate	C ₂ H ₆	
Corona PD	H ₂	Low	H ₂	VeryLow
S, T<200°C Aging	CO ₂ Furans, alcohols Low DPs of paper	Very Low		

C.4. Typical gas concentration levels in the DGA database of the WG versus type of fault

C.4.1. Typical concentration levels

90% percentile concentration values of gases are used as “typical” values by IEC and CIGRE and as “condition 1” values by IEEE, in order to concentrate maintenance efforts on the highest 10% of ppm values of each gas, where the corresponding transformers are potentially most at risk because of this higher gas formation.

Above typical values, gas formation is considered to be due to a fault by IEC/ CIGRE/ IEEE (see Terms and Definitions), and it is recommended to increase oil sampling frequency and perform additional monitoring tests when needed.

Below typical values, gas formation is not considered to be due to a fault by IEC/ CIGRE/ IEEE, but to a stress in the transformer (see Terms and Definitions).

The 90% percentile typical concentration values calculated on the whole DGA database of the WG are indicated in App Table C.5 ("Whole Database, Percentile 90"), App Table C.6 (last column) and App Table C.2 ("All results").

C.4.2. Typical concentration levels versus type of fault

The effect of type of faults on 90% percentile typical concentration values has been calculated in each DGA sub-database of the WG for each gas. An example of such calculations in Excel form is indicated in App Table C.5.

App Table C.5.: Example of Calculations of Percentile Values of Gas Concentrations vs Type of Fault or Stress according to DGA 14

Whole Database		Number of Cases	Percentilie	H2	CH4	C2H6	C2H4	C2H2	CO	CO2	O2	N2	
		337800	30	5	3	1	1,5	1	104	1168	3455	48067	
			60	20	12	9	9	1	280	2577	12400	66408	
			90	118	85	111	56	5	700	6300	27500	81343	
Fault Condition	Number of Cases	Percentilie	H2	CH4	C2H6	C2H4	C2H2	CO	CO2	O2	N2		
	D1	26483	30	7	3	1	1	1	98	900	6420	48347	
			60	25	5	2	1	1	229	1805	16326	65619	
			90	160	22	13	30	136	504	3980	28267	79700	
	T1	47164	Percentilie	H2	CH4	C2H6	C2H4	C2H2	CO	CO2	O2	N2	
			30	10	38	36	3	1	151	1820	1079	42754	
			60	31	68	104	7	1	326	3358	3911	65990	
				90	210	163	314	20	1	784	7820	11624	82838
	T2	28388	Percentilie	H2	CH4	C2H6	C2H4	C2H2	CO	CO2	O2	N2	
			30	13	25	14	13	1	203	2376	1148	53374	
			60	34	46	52	22	1	457	4400	3847	71237	
				90	180	151	246	75	1	909	9507	12965	86994
	T3	72989	Percentilie	H2	CH4	C2H6	C2H4	C2H2	CO	CO2	O2	N2	
			30	4	3	1	14	1	167	1801	7316	59565	
			60	16	9	6	37	1	367	3352	19000	68737	
				90	84	76	66	221	3	783	7668	29200	82514

Results of these calculations of typical values vs type of fault or stress in ppm in the DGA database of the WG are indicated in App Table C.6, column 5. Only gases which are significantly influenced by the type of fault or stress are indicated, in column 4.

Details on calculations made for App Table C.5 and App Table C.6 are indicated in sections C.2 and C.3.

App Table C.6. Typical Concentration Levels vs Type of Fault or Stress in the Subdatabases of the WG^{1,14}

Types of Faults or Stresses Identified by DGA In the Subdatabases Using:			90% Typical Concentration Levels in ppm Calculated from:		
Subdatabases:	Triangles:	Type of Fault or Stress :	Gas	Subdatabases (where Faults or Stresses were identified by DGA)	Whole Database (Faults or Stresses Unknown)
T2+T3	5	T3-H	C ₂ H ₄	126	56
		C		314	
T1+T2+PD	4	O	H ₂	550	111
		S		463	
		Corona PD		23,400	
D1	1	D1	C ₂ H ₂	8	5

Note: faults or stresses identified by DGA in App Table C.6 may not always be the actual faults found by inspection. However, in the majority of cases actual faults or stresses are relatively well predicted by DGA, thanks to the very large number of inspected DGA cases accumulated worldwide during the past 50 years. Furthermore, the vast majority of faults or stresses in transformers in service are identified by DGA, while only a very small number of them have been identified by visual inspection.

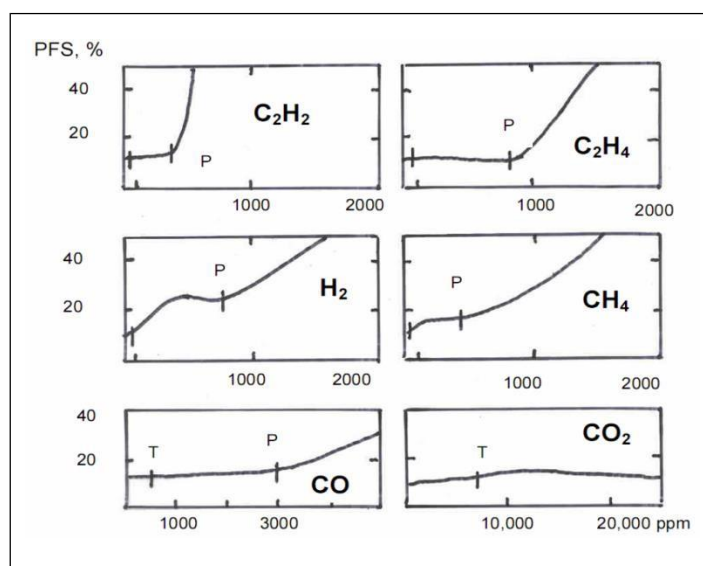
The same calculations were made on two larger, more uniform databases¹⁶ (mostly from utilities outside of the WG in the US). In the two databases the occurrence of faults or stresses in % and the 90% percentile typical values were relatively similar, confirming that the DGA database of the WG is as representative of DGA formation in transformers as larger, more uniform databases (see C.1.2).

C.5. Gas concentration levels above typical values

C.5.1. Gas concentration levels above typical values at CIGRE and IEC

Above typical values, “pre-failure” (PF) values have been used by CIGRE and IEC and published in TB 443 [B2], TB 296 [B1] and IEC 60599 [B3].

They are based on the calculated probabilities of having a failure-related event (PFS) (e.g., a Buchholz alarm or actual failure) as a function of gas concentration level, as indicated in App Figure C.1 [B23] for each gas. There is an inflexion point in these curves, above which the PFS increases rapidly, which has been defined as the prefailure (PF) value in the above CIGRE and IEC publications.



App Figure C.1. Probability of Having a Failure-Related Event (PFS) in % vs Gas Concentrations in ppm; T = Typical value; P = Prefailure value (PF)

As can be seen in App Figure C.1, below typical 90% percentile values (T), the PFS is low [B3] (typically, ~10% when using manual sampling), even though failure cannot be totally ruled out.

Above prefailure (PF) values, the PFS increases rapidly (up to 30 or 50%). Even though a failure may never occur above PF values, the likelihood of having one is higher [B3].

Prefailure (PF) values were calculated by several members of CIGRE TF D1.01/A2.11 [B1], following the method indicated in IEC 60599 [B3], and are indicated on the last line of App Table C.7. They were found to be surprisingly similar on different networks, and therefore were adopted in CIGRE TB 296 [B1] and 443 [B2].

The same PF values have been used by default in this TB, in App Table C.7 (they could not be recalculated with the DGA database of the WG because of missing information information such as failure-related events in the individual databases).

C.5.2. Gas concentration levels above typical values

App Table C.7. CIGRE Gas Concentration Levels above Typical Values, in ppm

Concentration Level	H ₂	CH ₄	C ₂ H ₄	C ₂ H ₆	C ₂ H ₂	CO	CO ₂
Typical	118	85	56	111	5	700	6300
Intermediate 1	200	135	120	210	19	970	11600
Intermediate 2	280	180	200	300	40	1180	16700
Pre-failure (PF)	725	400	800	900	450	2100	50000

In App Table C.7, the only small difference with TB 443 are typical values, which are those recalculated on the DGA database of the WG in App Table C.2 (“All results”) and App Table C.5 (“Whole Database, Percentile 90”). Intermediate values 1 and 2 have been recalculated by extrapolation following the method used in TB 443, based on oil sampling intervals required at these gas levels (monthly and weekly, respectively). An algorithm is available for doing these calculations¹.

Note 1: readers not comfortable with the calculated prefailure (PF) values of TB 443 can replace them in App Table C.7 by their own sets of values observed on average just before failure of their transformers.

Note 2: readers are encouraged to calculate their own 90% percentile typical gas concentration values (using their own DGA database), as recommended by the IEC, and to use these values in App Table C.7.

As in TB 443 and often asked¹⁵, if the concentration of only one gas in a transformer exceeds the concentration levels indicated in App Table C.7, then the transformer will be considered as being at this gas level. For example, if hydrogen is above 118 ppm, the transformer will be at intermediate 1 level. If at the same time methane is above 85 ppm, it will remain at the same intermediate 1 level.

C.5.3. Gas Concentration Levels Above Typical Values versus Type of Fault

In App Table C.7, the type of fault is not known. Values in App Table C.7, however, can be modified to take into account the type of fault.

For instance, in case the fault responsible for gas formation has been identified as T3-H, the most significant gas is ethylene, as indicated in Chapter 3.6.

The typical value of ethylene in App Table C.7 (56 ppm) can thus be replaced in the case of a fault T3-H by 126 ppm, as indicated in column 5 of App Table C.6.

The PF value of ethylene can then reasonably be assumed to be affected in the same way as the typical value. So the PF value of ethylene in App Table C.7 can be replaced by 800 ($126/56=1800$) ppm.

Intermediate values 1 and 2 can then be recalculated as 270 and 450 ppm, respectively.

These values are those indicated in Table 3.8 of Chapter 3 for faults T3-H.

Values in App Table C.7 for the other gases have been observed to remain unchanged in case of T3-H faults.

The same can be done for the other types of faults and gases in App Table C.6. An algorithm is available for these calculations. Values thus calculated for each type of fault are indicated in Table 3.1, Table 3.4, Table 3.8, Table 3.11; Table 3.14, Table 3.18, Table 3.22 and Table 3.25 of Chapter 3.

C.6. Gas concentration levels above typical values versus location of fault

Values in App Table C.7 can also be modified when more information about the location of fault is available.

For instance, in case of faults D1, failure has been observed to occur at very different levels of acetylene depending on whether the fault D1 is in oil or in paper (in oil, at 1400 ppm of acetylene or more; in paper at around 45 ppm or less, see sections 3.7 and 3.11 of Chapter 3).

These values can be used as more specific prefailure (PF) values in case of faults D1 in oil or in paper. The prefailure (PF) value of acetylene in App Table C.7 is 450 ppm when the type of fault is not known.

The typical value of acetylene in case of a fault D1 in App Table C.6 is 8 ppm.

Therefore, in case the fault D1 is in oil, the typical value of acetylene becomes $8 (1400/450)=25$ ppm.

In case it is in paper, it becomes $8 (45/450)=0.8$ ppm.

These are the values indicated in Table 3.11 and Table 3.26 of Chapter 3, respectively.

Similarly, case studies reported to the WG⁵ indicate that in cases of faults carbonization of paper C, failure occurred at: 2900 ppm or more of ethylene when the fault was in leads; 970 ppm when the fault was on the outside of windings; 25 ppm when the fault involved localized carbonization of paper between turns of one winding disk¹⁰ (see sections 3.8 to 3.10 of Chapter 3, respectively).

When the type of fault is not known, the prefailure (PF) value of ethylene in App Table C.7 is 800 ppm.

In case of a fault carbonization of paper C, App Table C.6 allows to calculate its PF value as: $800 (314/56)=4480$ ppm, its typical value being 314 ppm.

In case it is a fault carbonization of paper C in leads, its PF value is 2900 ppm as mentioned above, and its typical value becomes $314 (2900/4480) = 200$ ppm.

Values calculated above are indicated in Table 3.14 of Chapter 3.

Similar calculations for faults C on the outside of windings give a typical value of ethylene of $314 (970/4480)=68$ ppm, as indicated in Table 3.18. And between turns of one winding: $314 (25/4480)=2$ ppm, as indicated in Table 3.22.

All these calculations, as well as combinations of type and location of faults, can easily be made with the help of an algorithm¹ and following a “decision tree”.

C.7. Gassing rate levels above typical values

The CIGRE gassing rate levels of TB 443 [B2] have been used in App Table C.8.

The only small difference with TB 443 are the typical gassing rate levels in App Table C.8, which are those recalculated on the DGA database of the WG, following the method described in [B23], section 3-B, after cleaning up the DGA database of cases with missing or duplicated data.

App Table C.8. Gassing Rate Levels Using Manual Oil Sampling (in ppm/month) when Faults Were Not Identified by DGA

Rate Level	H ₂	CH ₄	C ₂ H ₄	C ₂ H ₆	C ₂ H ₂	CO	CO ₂
Typical	1.8	1.3	1	1.2	1	16	135
Intermediate 1	6	5	4	7	0	60	520
Intermediate 2	11	12	11	17	1	130	1140
Pre-failure (PF)	90	150	150	330	15	1420	12500

Modifications to values in App Table C.8 as a function of type and location of faults can be made as in the case of App Table C.7 if desired.

On-line monitors in general allow to calculate gassing rates much more reliably and frequently than manual oil sampling [B24]. Also, to more precisely calculate the gassing rates that occurred in transformers just before failure (PF rates). These prefailure (PF) gassing rates are usually much

higher than those which occurred days, weeks or months before failure when using manual sampling, which are more representative of intermediate or typical rates.

Prefailure (PF) to typical rates thus calculated with on-line monitors, e.g. in [B25], have been observed to be about 10 times higher than those calculated with manual oil sampling in App Table

C.8. A similarly very high prefailure (PF) rate level of 5 ppm per day of acetylene, and even of 5 ppm per hour in case of an abrupt spike, has also been used in Germany when using an on-line monitor [B26].

By contrast, typical to prefailure (PF) gas concentrations calculated with on-line monitors are relatively similar (with a few differences related to the databases used) to those calculated with manual oil sampling in App Table C.7, probably because gas concentrations are less affected by how close they are to failures.

C.8. Recommendations based on gas levels, type and location of faults

Recommendations based on gas levels, type and location of faults were among the tasks of WG D1/A2.47 (see Chapter 1, Introduction).

The recommendations of IEC 60599 [B3] have been used by the WG, based on gas levels taking into account the type and location of faults:

- Typical gas levels: “increase oil sampling intervals from yearly to monthly, consider performing complementary tests (acoustic, electrical, infrared) or reduce loading”.
- Intermediate 1 levels: “increase oil sampling to weekly”.
- Intermediate 2 levels: “consider installation of on-line multi-gas monitors^{10,17} and inspection depending on results of complementary tests”.
- Prefailure (alarmgas levels): “consider immediate action, removal from service for repair or replacement depending on damage observed”.

Note: these IEC recommendations based on gas levels (gas concentrations or gassing rates) are given for information only. Actual actions on the equipment are the sole responsibility of engineers in charge of the transformers.

Complementary acoustic emission (AE) tests [B27] mentioned above are particularly sensitive to active faults T3, T3-H or T2, since “boiling” of oil will occur at these temperatures, producing easily detectable acoustic emissions. Also in case of active sparking partial discharges (D1) and partial discharges of the corona-type PD.

If the faults are deeply within the windings or core, however, they are less likely to be detected by AE. The same with infrared cameras. AE sensors should be installed properly on transformer tanks in order to get reliable signals⁸.

Dielectric response analysis through the use of Polarization Depolarization Current (PDC) and Frequency Domain Spectroscopy (FDS) can also be used as a complementary test² [B3] to detect and locate:

- overheating in oil (T3): see APPENDIX F.3.
- moisture in pressboard or paper, which increases dielectric losses, carbon dioxide, also aging of paper,
- conduction at oil/paper interface.

In case there is some acetylene, winding resistance measurements can be made to assess any looseness of contacts which might develop into overheating or arcing².

See [B28] concerning the use of DGA for health indexes. It has been mentioned that they can be used for a coarse screening of DGA results, but will never replace human experts³⁵.

APPENDIX D. Typical concentration levels and faults in wind-farm transformers and bushings

Note: values in APPENDIX D are coming from databases outside of the WG.

D.1. Wind-farm transformers

Wind-farm transformers are well-known^{1,18,19} to generate high to very high levels of hydrogen due to stray gassing of oil S and partial discharges of the corona-type PD (up to 55,000 ppm in some cases). Also, other gases due to thermal faults T3.

This high gas formation, without failure in most cases, is due to the transformers being of the Padmount-or distribution-type used, which were not designed for the highly variable loads they are subjected to, creating harmonics and overheating. Also, due to tank constraints, they cannot be oil-impregnated under vacuum, resulting in the formation of voids in paper and partial discharges of the corona-type PD.

It is sometimes believed²⁰ that very high levels of hydrogen (several thousand or tens of thousands of ppm) in transformers cannot be due to stray gassing (overheating) of oil, but only to partial discharges of the corona-type PD. As indicated in APPENDIX G.2, however, numerous such cases were reported to the WG, which were not due to partial discharges of the corona-type PD (no PD signal during PD tests on the transformers³), but indeed to stray gassing of oil only.

Another common belief²⁰ is that extensive partial discharges of the corona-type PD in paper are always harmful to insulation because of the formation of x-wax. This is not supported by the numerous observations of transformers operating with tens of thousands of ppm of hydrogen without failure. If such levels were always harmful, the transformers would have failed well before reaching these high levels of hydrogen. So, partial discharges of the corona-type PD do not significantly affect the operation of transformers in the short term. If they evolve into more dangerous partial discharges of the sparking-type, they will immediately initiate the formation of acetylene, indicating the fault progression.

The 90% percentiles of gas concentrations formed in wind-farm transformers from various sources are indicated in App Table D.1:

App Table D.1. 90% percentiles of gas concentrations in ppm

Source		H ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	CO	CO ₂
Iberdrola ²⁹	Spain	1026	184	-	-	-	555	2825
Acciona ²¹	Spain	5400	788	145	8	0	789	1794
Alliant ³⁰	US	5660	344	53	15	1	521	1592
SDMyers ³⁴	US	2352	593	162	47	5	977	5173

Note: Iberdrola is using distribution transformers and Acciona padmount transformers²¹.

The relative proportion of faults observed in wind farm transformers of various sources, using methods in APPENDIX H is indicated in App Table D.2, in %:

App Table D.2. Relative proportion of faults, in %, in wind-farm transformers from various sources

Source		PD	S	O	C	T3	D
Iberdrola ²⁹	Spain	-	69	-	23	25	-
Acciona ²¹	N. America	24	74	-	-	1	-
Alliant ³¹	US	47	14	-	-	30	8
Martec ³¹	Zambia	11	41	44	-	13	-
Select Sol ³²	Australia	100	-	-	-	-	-
VonRoll ³³	Israel	100	-	-	-	-	-

Hydrogen-absorbing Jarylec additives [B29] can be used to reduce the amount of hydrogen (actually, ionized hydrogen) produced by partial discharges of the corona-type PD. However, they have no effect on the molecular hydrogen produced by stray gassing (overheating) of oil.

Since many wind farm transformers are installed off-shore, synthetic ester oils (e.g., in GB) and silicone oils (e.g., in Denmark and Belgium) are often used, as they are viewed as more fire-resistant and less damaging to the public image of their owners in case of oil spills. Stray gassing of silicone oils forms hydrogen, but only at temperatures > 200°C¹ (see APPENDIX G.2). For fault identification in ester and silicone oils, see APPENDIX H.8.

D.2. Bushings

95% typical concentrations of gases have been calculated in bushings of WG members at EDF³ (France), Energopomiar-Elektryka (EE)¹⁹ (Poland) and Siemens²⁰ (Germany), and are indicated in App Table D.3. These values do not take in consideration the type of fault.

They are significantly higher in some cases than those presently in IEC 60599 for bushings [B3] and reported in last line of App Table D.3, so they will be proposed as an addition to IEC 60599, after being submitted for approval by the TC on bushings.

In case of faults S or partial discharges of the corona-type PD, values of hydrogen of up to 70,000 ppm have been recently reported in bushings in Sweden and India with no failure [B30], [B31]. See also sections 3.2.2 and 3.3.

App Table D.3. 95% percentiles of gas concentrations in ppm in bushings in service at EDF (France), Energopomiar Elektriika (Poland) and in Germany

Bushing type	Source	H ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	CO	CO ₂	C ₃ H ₈	C ₃ H ₆
Oil/air-400kV	EDF ³	110	94	202	29	0	1203	7624	228	65
Oil/SF ₆ -400kV	EDF ³	536	553	272	132	1	1130	15,825	265	151
Oil	EE ¹⁹	563	67	72	11	8	708	5437	56	14
Oil-400kV	Germany ²⁰	170	52	22	3	2	249	525	12	3
Oil-<400kV	Germany ²⁰	335	113	45	7	1	311	1142	19	5
Oil	IEC 60599	140	40	70	30	2	1000	3400	-	-

Note: results at EDF are based on 420 oil/air bushings and 256 oil/sulphur hexafluoride bushings.

APPENDIX E. Use of carbon monoxide and carbon dioxide to detect faults in paper insulation

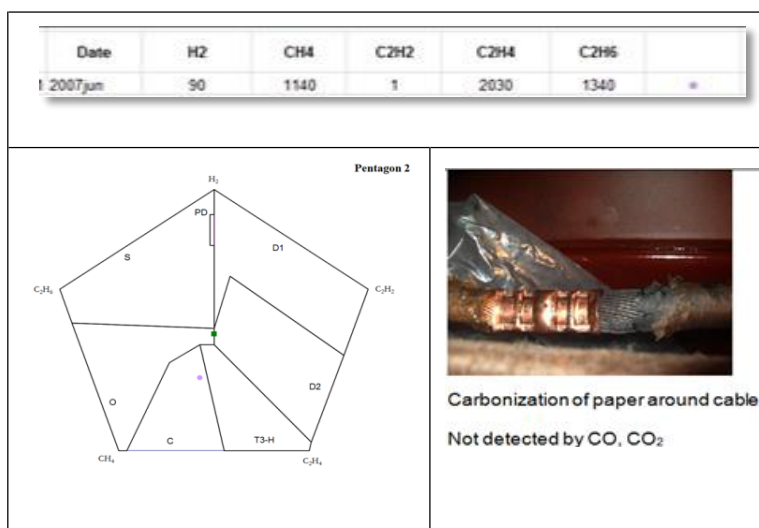
E.1. Use of carbon monoxide and furans for detecting carbonization of paper

High values of carbon monoxide (> 1000 ppm) and low values of the carbon dioxide/carbon monoxide ratio (< 3), in the presence of significant amounts of the other hydrocarbon gases and hydrogen and furans, are generally considered as a confirmation of a fault in paper with carbonization of paper C. However, they should be used with caution, as in many cases values of carbon monoxide and carbon dioxide may be misinterpreted.

For instance, a general consensus today is that high values of carbon monoxide (> 1000 ppm) and/or low values of the carbon dioxide/carbon monoxide ratio (< 3), in the absence of significant amounts (e.g., $< a few$ ppm) of the other hydrocarbon gases and hydrogen, are **NOT** an indication of a fault C with carbonization of paper, but rather of oil oxidation under conditions of low oxygen content in the oil (e.g., oxygen/nitrogen ratio < 0.2 , see APPENDIX F)^{1,20,22}, as indicated in the latest edition of IEC 60599.

This has been reported by many members of the WG worldwide (in Croatia, Serbia, Germany, Finland, Russia, US, etc), particularly in the new, more compact transformers of the sealed-type available on the market in the last decade, which tend to slightly overheat and oxidize oil more.

Many other observations reported by WG members²³ and in [B8] also indicate that faults with carbonization of paper C of relatively small volumes (e.g., on leads or turns) may not generate large enough amounts of carbon monoxide (or furans) to be detectable against the usually large background of this gas in oil. An example of a fault C in paper not detected by carbon monoxide and carbon dioxide but by the hydrocarbon gases and Pentagon 2 is indicated in App Figure E.1.



App Figure E.1. Fault with Localized Carbonization of Paper C, not Detected by CO, CO₂ but by Hydrocarbons Gases (i.e., in zone C of Pentagon 2)²³

High concentrations of carbon monoxide therefore do not necessarily mean that there is a fault in paper, while no significant increase in carbon monoxide necessarily means that there is not a fault in paper [B8]. The interpretation of carbon monoxide should thus be done with a lot of caution.

On the other hand, sudden increases in carbon monoxide and decreases in the carbon dioxide/carbon monoxide ratio < 3 , together with significant increases of the hydrocarbon gases, reinforce the possibility of a fault in paper C^{1,20}.

E.2. Use of carbon dioxide and furans to detect low temperature faults (> 130°C) in paper

Faults (above typical values) or stresses (below typical values) of type T1 < 300°C or O < 250°C in paper do not produce carbonization of paper but only some degradation of it (e.g., paper turning brown) and a decrease of its degree of polymerization (DP).

Overheating of paper above the hottest spot temperatures in windings specified for transformers in the loading guides of IEC and IEEE, typically 105°C under nominal load and 130°C during short periods of overload, is not considered as normal ageing of paper but as a fault or stress and is therefore within the scope of the WG.

A good correlation between high cumulative values of carbon dioxide (> 15,000 ppm) and carbon dioxide/carbon monoxide ratio (>20) and low average DP of paper (< 400) in transformers has been reported in TB 296 [B1] and 443 [B2]. “Cumulative” means that values observed before degassing the oil, when this had been done, have been added to values observed after degassing.

Carbon dioxide/carbon monoxide ratios of 15 to 50 have thus been observed [B1] on transformer laboratory models with simulated fault temperatures of 160°C to 125°C, respectively, and confirmed in full-size transformers in service. This has also been confirmed by laboratory tests done recently in the WG² with both Kraft and thermally upgraded paper at 100° C and 150° C.

It can also be seen in App Figure C.1 that the probability of having a failure-related event (PFS) in transformers does not significantly increase at very high levels of carbon dioxide. This suggests that the PFS also should not increase significantly at low DPs of paper.

This is supported by the large number (~130+) of transformers surveyed by members of the WG, with DPs of paper between 250 and 100 and therefore with a low tensile strength of paper, still operating normally without failure, even when subjected in some cases to external short circuits or transported. This is also supported by the DP limit of 150 used in China²⁴.

Most low DP values in this survey were detected first by high values of carbon dioxide or furans, or by visual inspection of highly degraded paper then confirmed by direct measurements on paper samples taken from the inspected transformers.

This suggests that a high tensile strength of paper is not needed in windings, but rather a good resistance to compression, which even very short cellulose fibers have, since windings are very rigid²⁰ and will hold cellulose fibers together between turns, even at very low clamping pressures and DPs of paper. See section 3.5 for examples.

Operation of transformers with these low DPs, however, requires that oil, which serves as a binder between the short cellulose fibers on the vertical sides of turns, is not removed from the tank²¹.

It is also very important to properly maintain the quality of oil throughout the life of transformers. Measurements of paper conductivity in the laboratory²⁵ show that there is no difference in conductivity between paper with a DP of 1132 and one with a DP of 188. By contrast, the conductivity of paper impregnated with highly oxidized oil has been observed to increase dramatically by a factor of 500 [B11], with dielectric failure of paper becoming much more likely than its mechanical failure due to its low DP.

APPENDIX F. Influence of the oxygen/nitrogen ratio on typical values

F.1. Use of the oxygen/nitrogen ratio

A decrease in the oxygen content or the oxygen/nitrogen ratio in transformers indicates oil oxidation because of oil overheating, which may be normal or abnormal. An increase in the oxygen content or the oxygen/nitrogen ratio in sealed transformers indicates a leak in the air-preservation system (nitrogen blanket or membrane) [B3].

The normal value of the oxygen/nitrogen ratio depends on the type of transformer. It has thus been calculated¹ that all nitrogen-blanketed transformers and about 60% of the membrane-sealed types in DGA databases have an oxygen/nitrogen ratio ≤ 0.2 , while all air breathing transformers and about 40% of the membrane-sealed types have an oxygen/nitrogen ratio > 0.2 .

This can be explained by the fact that membrane-sealed transformers are initially filled with air-containing oil which has been only partially degassed and which therefore has a high oxygen/nitrogen ratio. Then, oxygen in them slowly decreases with time because of oil oxidation and they move to the low ratio category¹.

F.2. Influence of the oxygen/nitrogen ratio on typical values in the DGA database of the WG 57

It has been observed at IEEE¹⁶ that the oxygen/nitrogen ratio has a significant influence on calculated 90% percentile typical values.

The influence of the oxygen/nitrogen ratio on 90% percentile typical values calculated from the DGA database of the WG is indicated in App Table F.1. It can be seen that at low oxygen/nitrogen ratios the typical values of hydrogen, methane and ethane (the low temperature gases) are higher.

App Table F.1. Influence of Oxygen/Nitrogen Ratio on Typical Values in the DGA Database of the WG

O ₂ /N ₂ ratio	Number of cases	H ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	CO	CO ₂	O ₂	N ₂
< 0.05	67,783	196	138	224	52	1	835	8481	3019	92,000
< 0.2	158,479	160	112	178	51	1	858	7790	10,827	87,200
> 0.05	270,017	101	65	80	57	8	658	5736	28,699	78,200
> 0.2	179,321	87	48	48	61	14	529	4870	30,472	75,800

F.3. Influence of the oxygen/nitrogen ratio on the occurrence of faults or stresses in the DGA database of the WG

This ratio has also been observed^{1,14} to have an influence also on the occurrence of faults or stresses (see App Table C.3). This is indicated in App Table F.2. As can be seen, more low-temperature faults or stresses T1, O, S, C and T2 tend to occur at low oxygen/nitrogen ratios (< 0.2) than at high ratios (> 0.2), and the reverse for high-temperature faults or stresses T3. It is not clear, however, whether low temperature faults or stresses are the consequence or the cause of the low oxygen/nitrogen ratios (through increased oil oxidation).

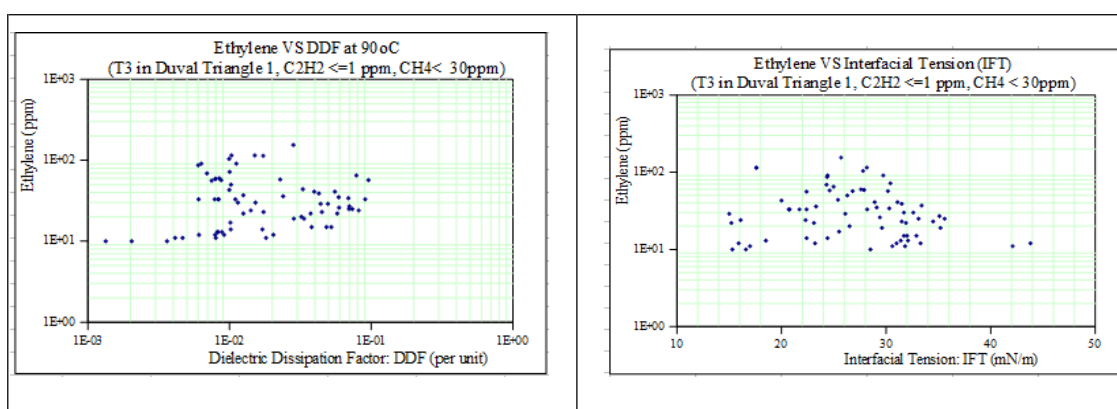
App Table F.2. Influence of Oxygen/Nitrogen Ratio on the Occurrence of Faults in the Database of the WG

O ₂ /N ₂ ratio in %	Number of cases	T3-H	T3	T2	T1	C	O	S	PD
< 0.05	67,783	19	15	19	29	9	15	18	0.04
< 0.2	158,479	19	18	14	23	7	12	13	0.04
> 0.05	270,017	21	23	6	10	6	5	5	0.04
> 0.2	179,321	22	25	4	6	5	5	4	0.04
All	337,800	21	22	8	14	6	7	8	0.04

Note: large power transformers, eg., GSU, which tend to be sealed and operate with a lower oxygen/nitrogen ratio, have indeed been reported¹³ to overheat more. The same is often observed in new transformers of compact design¹⁹.

Some high oxygen/nitrogen ratios in App Table F.1 and App Table F.2 in New Zealand are from free-breathing or sealed type transformers with deteriorated preservation system where oxidation of oil occur [B39]. Oxidation by-products increase the dielectric dissipation factor (DDF) & the dielectric losses and decrease the interfacial tension (IFT), producing oil overheating T3.

In this case, ethylene is the dominant gas and in many cases it is not accompanied by any methane or ethane. The two charts below, of ethylene vs. DDF and ethylene vs. IFT in App Figure F.1 are from 75 free-breathing transformers² in the DGA database of the WG. Further explanations about the correlations in App Figure F.1 can be found in [B39].

**App Figure F.1. Ethylene versus DDF and IFT of Oil with High Oxygen/Nitrogen Ratios²**

In Poland, low temperature stress or fault in old transformers are often associated to oil oxidation (high $\tan \delta$ and NV, low resistivity and IT). These oils can still be transparent but already give a sludge when filtered (e.g., for counting particles). In such cases, ethane is the dominant gas and always predominate on ethylene¹⁹.

APPENDIX G. Stray gassing of oil

G.1. Definition of stray gassing of oil

Stray gassing of oil has been defined in CIGRE TB 296 [B1] as the formation of gases in oil heated at temperatures between 90°C and 200°C somewhere in the transformer.

Stray gassing of oil should not be confused with the “Gassing Tendency of Oils Under Electrical Stress and Ionization”, which occurs when a gas phase above the oil is subjected to corona partial discharges of the corona-type PD, and which is measured with IEC 60628-A [B9] and ASTM D 2300 [B10].

G.2. Detection and significance of stray gassing in transformers

Faults or stresses stray gassing of oil S in transformers can be detected by Triangles 4 and by Pentagons 1 and 2. In the very large majority of oils and transformers, they are of the “usual” stray gassing type described in G.3.

Stray gassing of oil in transformers is of little concern because the temperature at which stray gassing of oil occurs (120° to 200°C) is not high enough to damage the electrical insulating properties of oil. Adjacent paper insulation will be at a much lower temperature and even less affected.

Indeed, very high levels of hydrogen have been reported in some transformers (e.g., 4800 ppm and up to 55,000 ppm, see Chapter 3.2), particularly in wind farm transformers and bushings (see APPENDIXES D.1. and D.2.) but also in power transformers, which were shown not to be due to partial discharges of the corona-type PD (no PD signal during PD tests), and which therefore were due to stray gassing of oil only, without affecting their normal operation and without resulting in failures^{1,19,26}.

Although stray gases do not affect the normal operation of transformers, they should not be considered as “irrelevant” gases that are due only to the chemical instability of oils and unrelated to what is happening in the transformers. They are rather an indication of abnormal oil temperature somewhere inside the transformer (not supposed to exceed ~105°C in transformer loading guides), due to design or operational problems, which may or may not need to be fixed after discussions with the transformer manufacturer. Stray gassing of oil may actually be viewed as a way to detect such abnormal overheating rather than being a nuisance¹.

Stray gases formed with some oils at 120°C occur sometimes very close to the boundary between zones S (stray gassing of oil) and PD (partial discharges of the corona-type) of Triangles 4 and Pentagons 1-2. In such cases, it is recommended to perform a stray gassing test in the laboratory on a sample of oil taken from the transformer, to verify that gas formation in it may indeed be due to stray gassing of oil. A PD test on the whole transformer may also confirm the absence of partial discharges of the corona-type PD.

G.3. “Usual” stray gassing of mineral oil

Around 20 different types of mineral oils were tested for stray gassing in 2005 by CIGRE TF D1.01/A2.11 in TB 296 [B1], and 100 by Doble in 2012. All produced mostly hydrogen at 120°C, and mostly ethane and methane at 200°C.

The amounts of stray gases formed have been reported to increase:

- With temperature. The same amounts have thus been observed during one laboratory test after 1 week at 120°C and after 30 weeks at 40°C²⁰. This, however, has not been reconfirmed nor observed in actual transformers.
- With high oxygen content in oil in the absence of inhibitors during laboratory tests and in free breathing transformers²⁰. Other reports, however, indicate that large amounts of stray gases (e.g., 600 ppm of hydrogen) have also been observed in membrane-sealed transformers¹⁰.
- A new stray gassing test in the laboratory using vials equipped with air-permeable septa has thus been proposed to better simulate stray gassing under transformer operating conditions, such as high oxygen in oil¹⁷. However, hydrocarbon gases and hydrogen formed in such vials have been observed to also easily diffuse to the atmosphere¹⁷, so how this may affect or not the results of such tests will have to be verified. Furthermore, it has been clearly demonstrated in Brazil [B33] that screw caps and septa of vials can produce ethylene and other gases at temperatures above 90°C, which will make the stray gassing test results totally unreliable. It will have to be verified if this occurs also at lower temperatures and during very long test durations (weeks or months). In such a case, glass syringes or sealable glass ampoules, as recommended in TB 296 [B1], may be more appropriate.
- It is to be noted that no stray gassing test can exactly simulate stray gas formation in transformers. Like oxidation resistance tests, they should be used mainly to compare and classify different oils.
- In the presence of some passivator additives in oil, particularly Irgamet 30 but also Irgamet 39. However, other reports show no significant increases.
- In the presence of copper and core steel in oil during laboratory tests^{19,20}, methane and ethane are produced in addition to hydrogen.
- In the absence of inhibitor additives in oil²⁰.

New oils on the market (e.g., “Gas-To-Liquid” GTL oils) may have different stray gassing patterns, which need to be determined precisely with stray gassing tests in the laboratory in order to identify them correctly in transformers in service. Recent tests suggest that the stray gassing patterns of GTL oils at 160°C are significantly different from those of conventional naphthenic oils [B32].

G.4. “Unusual” stray gassing of mineral oil

The unusual formation of ethylene as a result of stray gassing of oil has been reported in a very few occurrences, as a result of:

- Decomposition of some passivator additives used in oil (e.g., Irgamet 39)²⁰.
- Reclamation of oil on aluminosilicates or bauxite¹⁰ to remove corrosive sulphur²⁷. Such absorbants are catalysts for oil cracking and may create instabilities in oils.
- Sometimes after reclaiming the oil on Fuller’s Earth or bauxite¹⁰ under high oxygen content.
- In some older, solvent based oils (e.g., Voltesso 36) not used any more¹.
- Screw caps used for vials (Brazil) [B33].
- Aged, humid oils, once considered a possible reason¹², were ruled out by other tests and observations.

In summary, ethylene may exceptionally be formed by stray gassing of oil under very special conditions. Stray gassing of oil in transformers, however, in the very large majority of cases is of the “usual” type^{1,20}.

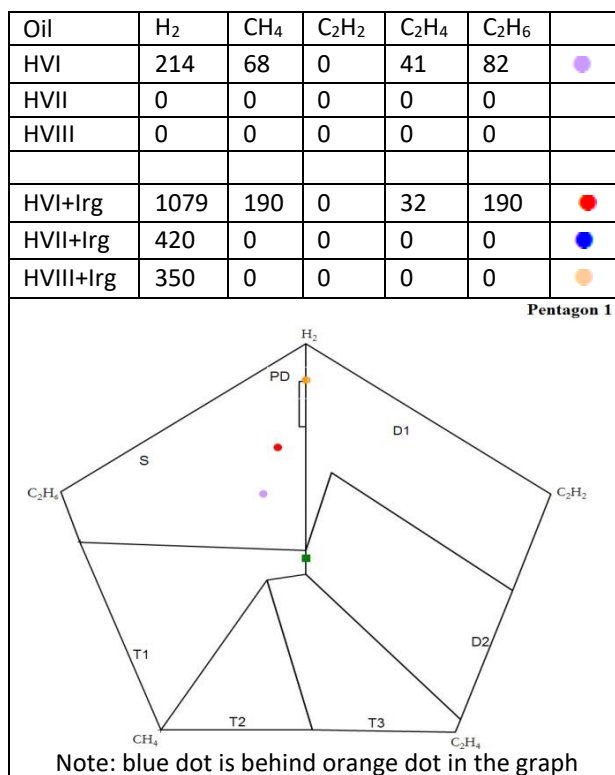
G.5. Stray gassing tests on mineral oils by the WG

A summary of stray gassing tests on mineral oils performed by one laboratory of the WG is indicated below¹⁸.

Commercial oils HVI, HVII and HVIII, meeting the requirements of IEC 60296 for standard oils, ASTM Type II oils and IEC 60296 for special applications, respectively, were used for these tests. All 3 oils were produced from the same refinery base stock and were subjected to the same stray gassing test described in CIGRE TB 296 (2010), in glass syringes, during 16 hrs at 120°C. Irgamet 39 passivator additive (Irg) was added to some samples (100 ppm). Results are indicated in App Table G.1. Gas values indicated as zero ppm mean that they were below the detection limits of the laboratory.

It can be seen that the more refined oils HVII and HVIII do not produce any stray gas, compared to the lesser refined HVI, which unexpectedly also produces some ethylene. Also that the addition of Irgamet 39 (Irg) results in larger amounts of stray gases formed, particularly hydrogen. Gas formation of all three oils in App Table G.1 occurs as expected in zone S of Pentagon 1.

App Table G.1. Stray Gas Formation from Commercial Oils HV¹⁸



A summary of stray gassing tests performed by another laboratory of the WG [B34], from 70°C to 200°C, with and without 0.3% of BHT (butylated hydroxy toluene) antioxidant additive, and using syringes and sealable glass ampoules, is indicated in App Table G.2.

It can be seen that gas formation moves as a function of temperature from zone O to zone T2 with uninhibited oils (without BHT additive), and from zone S to zone T2 with inhibited ones. The amounts of gases formed, however, are relatively similar, although a bit lower in the presence of BHT.

It is to be noted that in a transformer, the same gas formation will be produced by a hot spot temperature higher than during the laboratory test, because of oil convection around the hot spot and rapid cooling of it with cooler surrounding oil. For instance, gas formation at 90°C in the laboratory will likely be produced by a hot spot of e.g., 130°C in the transformer.

App Table G.2. Stray Gas Formation between 70°C and 200°C with and without BHT Additive

ID #	Date	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	
no bht	70c	6	9	0	0	10	●
	95c	7	26	0	0	12	●
	120c	14	42	0	0	13	●
	150c	9	56	0	37	14	●
	200c	72	160	0	72	63	●
ID #	Date	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	
bht	70c	5	2	0	0	0	●
	95c	9	4	0	0	0	●
	120c	23	15	0	0	2	●
	150c	9	35	0	19	9	●
	200c	82	170	0	90	55	●

Pentagon 1

Pentagon 1

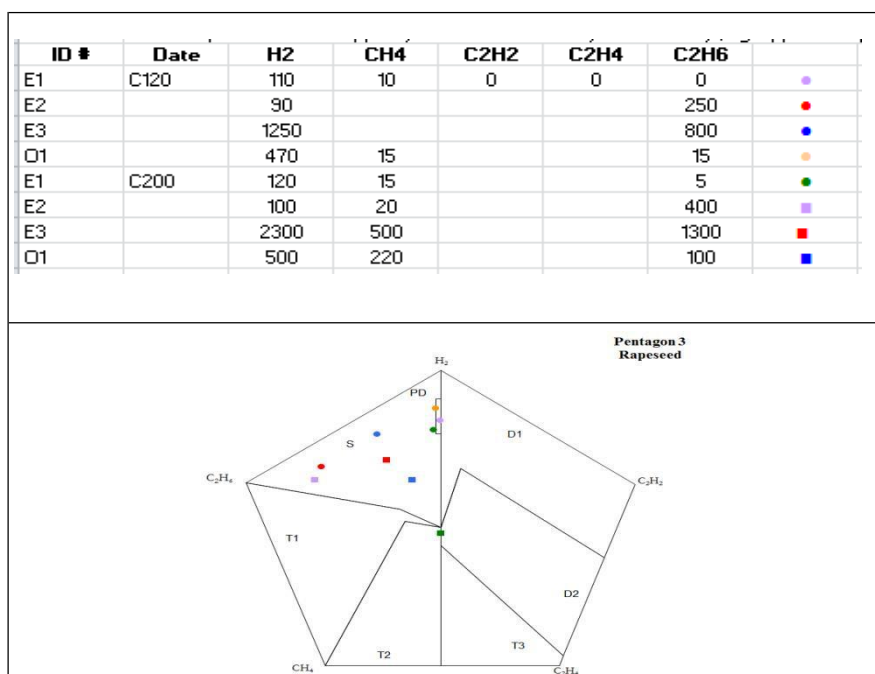
G.6. Stray gassing of non-mineral oils in pentagons 3

While stray gassing of mineral oils at 120 °C produces mainly hydrogen, stray gassing of ester oils produces mainly ethane^{1,15,20}.

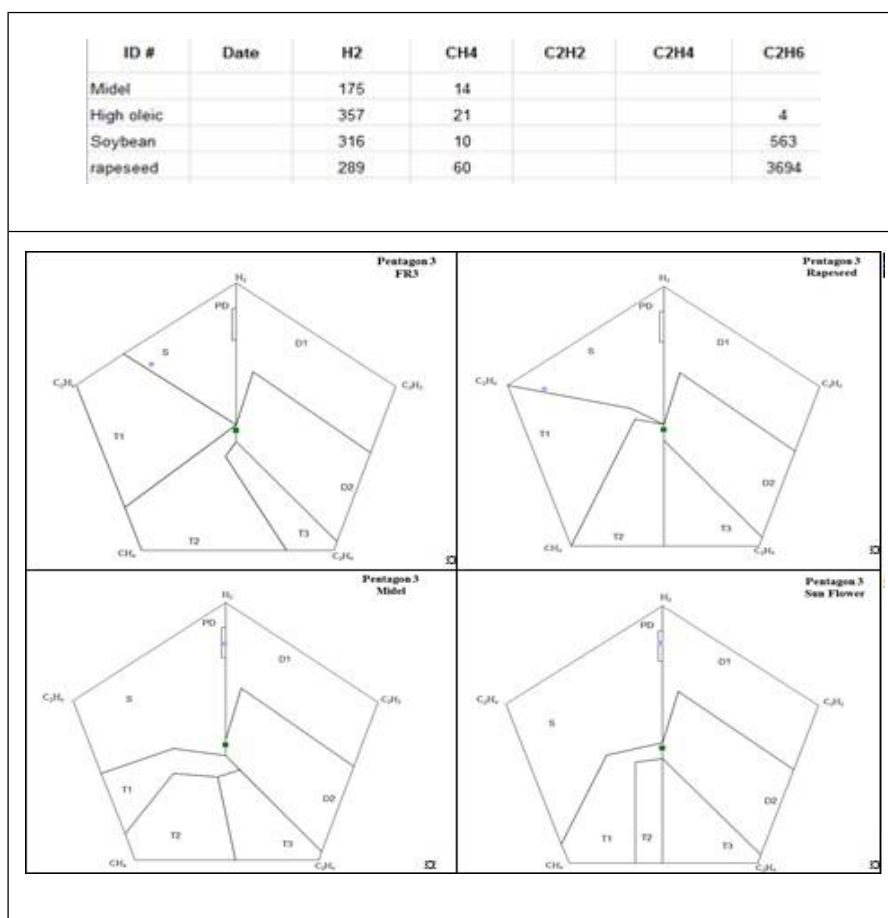
There are significant differences in the stray gassing patterns and shapes of stray gassing zones S in Pentagons 3 depending on the various types of natural and synthetic esters^{1, 13}, as can be seen in App Table G.3.

Stray gassing of silicone oils produces only hydrogen, but at much higher temperatures of 180 to 220°C.

App Table G.3 indicates the results of three different types of ester oils (E1-E3) and one mineral oil (O1) tested for stray gassing in the laboratory at 120 and 200°C in Italy¹⁵. Pentagon 3 for rapeseed oil has been used by default in App Table G.3 since the type of ester oils was not provided.

App Table G.3. Stray Gassing Tests with Ester Oils in Italy¹⁵

App Table G.4 indicates the results of four different types of ester oils tested for stray gassing at 120°C in the US¹³. App Table G.4 suggests that additional stray gassing tests at 200°C would be needed to precisely determine the stray gassing zones S of Pentagons 3, presently based mostly on pyrolysis tests.

App Table G.4. Stray Gassing Tests with Ester Oils in the US¹³

G.7. Stray gassing of oil in international standards

Stray gassing of oil is not a functional requirement preventing transformers from operating normally (see G.2), but it sometimes makes the interpretation of DGA in transformers a bit more difficult.

IEC Technical Committees TC14 and TC10 have therefore proposed to put mandatory limits on the amounts of stray gases formed in new unused oils at 105°C.

This, however, may unduly reject some oils which otherwise would be totally acceptable in terms of functional properties (electrical insulation, cooling, etc). Also it may result in litigations between transformer users and manufacturers, since stray gas formation in service above 105°C may be attributed wrongly to a transformer design problem rather than to the oil.

Acceptable limits of stray gas formation in oils on the market should therefore be left to oil users, not to standardization bodies to decide.

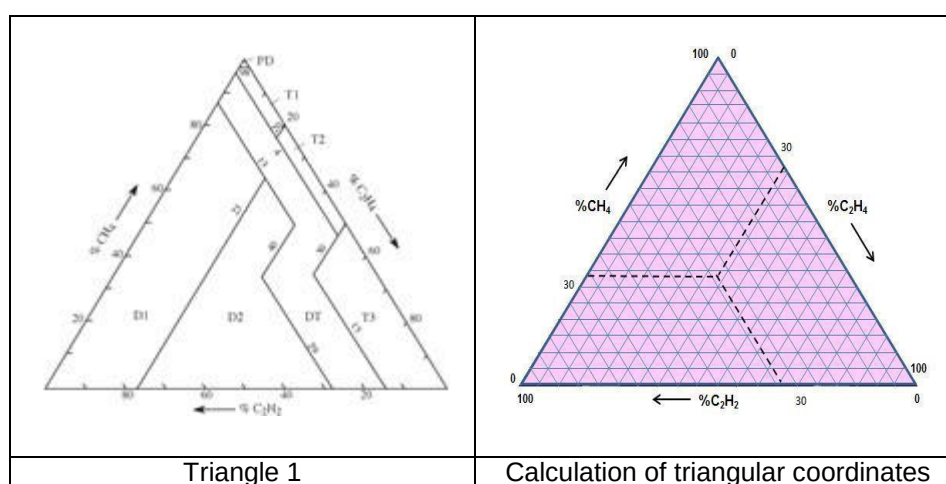
Furthermore, all oils in present transformers are more or less stray gassing, so the need to distinguish stray gassing from partial discharges of the corona-type PD will remain for a very long time, using for example the triangle and pentagon methods, whatever the new oils on the market.

APPENDIX H. How to draw Duval triangles and pentagons

Note: see Chapter 2.2 for minimum concentration values for using the Triangles and Pentagons

H.1. Duval triangle 1 for mineral oils

Triangle 1 for mineral oils (App Figure H.1) uses three gases corresponding to the increasing energy content or temperature of faults: methane (CH_4) for low energy/temperature faults, ethylene (C_2H_4) for high temperature faults, and acetylene (C_2H_2) for very high temperature/energy/arcing faults. On each side of the Triangle are plotted the relative percentages in % of these three gases.



App Figure H.1. Duval Triangle 1 and calculation of triangular coordinates

Triangle 1 allows identification of the 6 basic types of single faults indicated in Table 2.1 of Chapter 2, plus mixtures of electrical/thermal faults in zone DT.

The procedure for calculating DGA points in Triangle 1 using triangular coordinates is indicated in App Figure H.1 (graph on the right). If for example, $\text{CH}_4 = \text{C}_2\text{H}_4 = \text{C}_2\text{H}_2 = 100$ ppm, first calculate $\Sigma = (\text{CH}_4 + \text{C}_2\text{H}_4 + \text{C}_2\text{H}_2) = 300$ ppm, then the relative percentage of each gas: $\text{CH}_4/\Sigma = \text{C}_2\text{H}_4/\Sigma = \text{C}_2\text{H}_2/\Sigma = 33.3\%$ in this example. This provides only one DGA point in the middle of the Triangle.

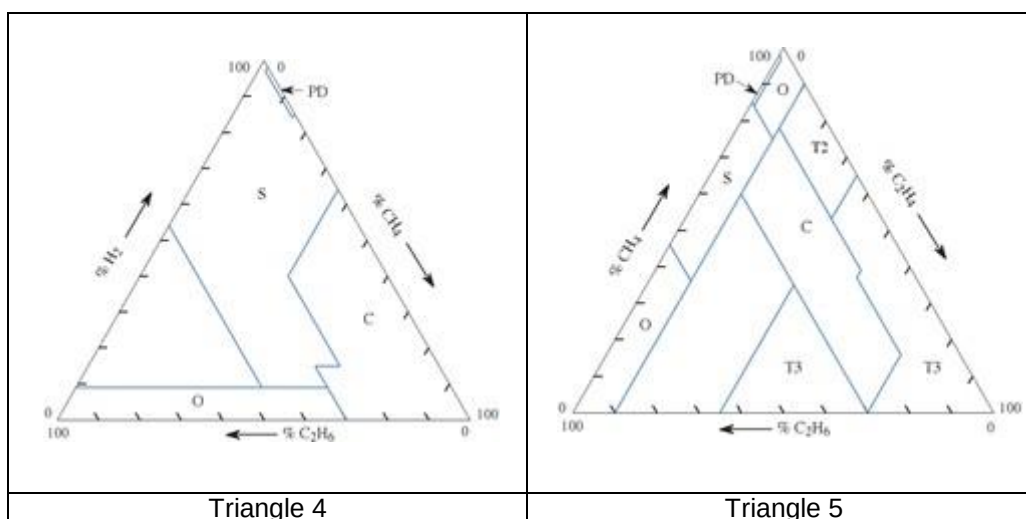
The same procedure is used for Triangles 4, 5 in App Figure H.2, modified Triangle 1 in App Figure H.5, and PEM Triangle in App Figure I.2.

Free algorithms for using the Duval Triangles Methods are available¹.

The advantages of the Triangle methods are that they always propose a fault identification (they are “closed” systems as compared to two-gas ratios methods), with few erroneous diagnosis (they are based on a large number of inspected cases of faulty transformers in service). They also allow to visually and rapidly follow the evolution of faults with time in a transformer.

H.2. Triangles 4 and 5 for mineral oils

When low energy or low temperature faults are identified using Triangle 1 (PD, T1 or T2), more information can be obtained on these faults with Triangle 4 for mineral oils (App Figure H.2), allowing to distinguish between faults S, O, PD, which are of relatively minor concern in transformers, and potentially more dangerous faults C.



App Figure H.2. Duval Triangles 4 and 5

When high, or very high, temperature faults have been identified with Triangle 1 (T2 or T3), more information can be obtained on these faults with Triangle 5 (App Figure H.2), allowing to distinguish between high temperature faults T3/T2 in oil only (T3-H), of lesser concern in transformers, and potentially more dangerous faults C involving possible carbonization of paper.

Triangle 4 should not be used, and Triangle 5 used only with caution, for faults identified as electrical faults D1 or D2.

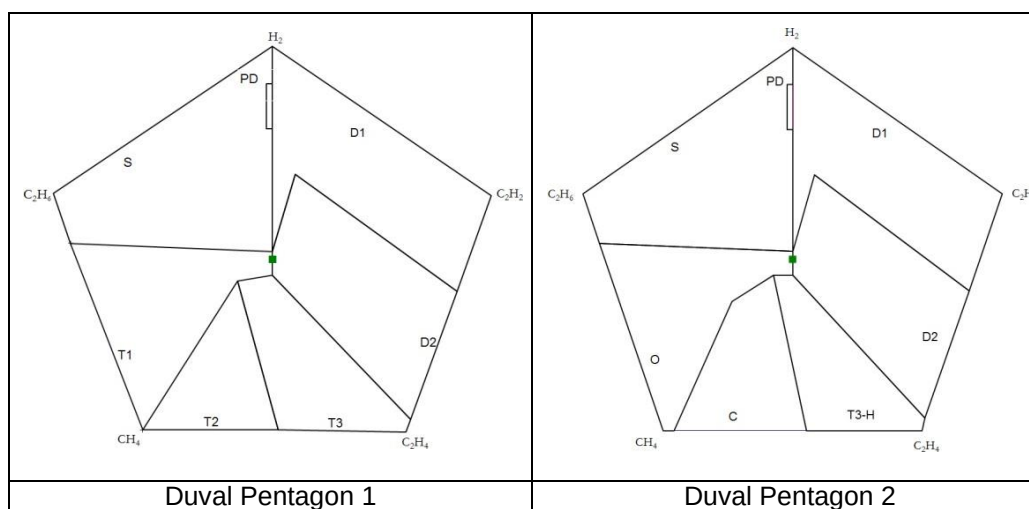
DGA points occurring in zones C of Triangles 5 and 4 indicate a possibility of carbonization of paper, not a 100% certainty, and require further investigations with carbon oxides and furans.

The unnamed zone between zones S and O in Triangle 4 is probably a zone of overheating between 200°C (fault S) and 250°C (fault O), although there are no inspected cases or laboratory tests to support that yet. Similarly, the unnamed zone between zones O, S, C and T3 of Triangle 5 is probably a fault zone of intermediate temperature.

H.3. Duval pentagons 1 and 2 for mineral oils

Duval Pentagons 1 and 2 (App Figure H.3) use the five hydrocarbon gases (hydrogen, ethane, methane, ethylene and acetylene). The order of gases at the five summits (apexes) of Pentagons 1 and 2 correspond to the increasing energy or temperature of the faults producing these gases (from hydrogen to acetylene).

Coordinates in the pentagons are Cartesian (x,y), starting from the center (green square) of the pentagons (see App Table H.2).



App Figure H.3. Duval Pentagons 1 and 2

The 6 basic types of faults of (PD, D1, D2, T1, T2 and T3) indicated in Table 2.1 of Chapter 2 can be detected with Pentagon 1, as in the case of Triangle 1, and also stray gassing of oil S.

If thermal faults (T1, T2, T3) have been identified with Pentagon 1, more information can be obtained on these faults with Pentagon 2, as in the case of Triangles 4 and 5.

Pentagon 2 allows to detect the three basic types of electrical faults (PD, D1 and D2) as in Pentagon 1, and to further distinguish between the four additional sub-types of thermal faults of Table 2.1 of Chapter 2 (S, O, C and T3 in oil only). In Pentagon 2, faults T3 in oil only are indicated as T3-H, where H is for Huile, which is the word for Oil in French.

DGA points occurring in zone C of Pentagon 2 indicate a possibility of carbonization of paper, not a 100% certainty, and require further investigations with carbon oxides and furans.

Free algorithms for using the Duval Pentagons Methods are available¹

H.4. When to use the triangles and pentagons?

If interest is only in the six basic types of faults of Table 2.1 of Chapter 2 (T3, T2, T1, D2, D1, DT, PD) and by single faults, DGA points can be displayed first in Pentagon 1 or Triangle 1.

If there is also an interest in the additional sub-types of thermal faults of Table 2.1 (S, O, C, T3-H), Pentagon 2 and Triangles 4 or 5 can then be used.

Finally, if there is an interest in detecting multiple faults (see APPENDIX H.6 and H.7), diagnosis provided by the Pentagons and the Triangles should be compared. If they do not agree, this may be an indication of multiple faults. Use subtracted (delta) values to further identify these multiple faults.

H.5. Numerical values for the triangles and pentagons

Numerical values for fault zone boundaries of Triangles 1, 4, 5 and Pentagons 1, 2 are indicated in App Table H.1 and App Table H.2, respectively.

App Table H.1. Numerical Values for Zone boundaries in Triangles 1, 4, 5

Triangle 1				Triangle 4				Triangle 5			
Fault	%CH ₄	%C ₂ H ₄	%C ₂ H ₂	Fault	%H ₂	%CH ₄	%C ₂ H ₆	Fault	%CH ₄	%C ₂ H ₄	%C ₂ H ₆
PD	98	-	-	PD	-	2,15	1	T3	-	35,50, 70,14	- ,35.30
T1	98	20	4	S	9,15	2,15,3 6	1,24,3 0,46	T2	-	10,35	12
T2	-	20,50	4	O	9	-	30	C	-	10,50, 70	12,14, 30
T3	-	50	15	C	15	36	24,30	O	-	10,34	- ,1,10, 2,14
DT	-	50,40	4,13,15 ,29					S	-	10	14,34
D2	-	23,40	13, 29					PD	-	1	2,14
D1	-	23	13								

App Table H.2. Numerical Values for Zone Boundaries of the Pentagons

Fault	(x,y) coordinates							
PD	0,33	-1,33	-1,24.5	0,24.5				
D1	0,40	38,12	32,-6,1	4,16	0,1.5			
D2	4,16	32,-6.1	24.3,-30	0,-3	0,1.5			
T3	0,-3	24.3,-30	23.5,-32.4	1,-32	-6,-4			
T2	-6,-4	1,-32.4	-22.5,-32.4					
T1	-6,-4	-22.5,-32.4	-23.5,-32.4	-35.3,3	0,1.5	0,-3		
S	0,1.5	-35,3.1	-38,12.4	0,40	0,33	-1,33.	-1,24.5	0,24.5
T3-H	0,-3	24.3,-30	23.5,-32.4	2.5,-32.4	-3.5,-3			
C	-3.5,-3	2.5,-32.4	-21.5,-32.4	-11,-8				
O	-3.5,-3	-11,-8	-21.5,-32.4	-23.5,-32.4	-35,-3.1	0,1.5	0,-3	

H.6. Detection of multiple faults or stresses in mineral oils

Triangles 1, 4, 5 and Pentagons 1, 2, as well as all other diagnosis methods (e.g., Rogers or IEC) were initially developed for detecting single faults only.

However, multiple faults (mixtures of faults) often occur in the same transformer rather than single faults. For instance, loose contacts may overheat first then produce sparking, or the reverse. Such multiple faults may be more difficult to identify with certainty [B5]. For instance, actual mixtures of faults T3+D1 or T3+D2 or T2+D2 may sometimes appear in terms of gas formation as faults D2 in Triangle 1, Pentagon 1 and other diagnosis methods (Rogers, etc.), while actual mixtures of faults T3 in oil (T3-H) and O may appear as faults C in Triangle 5 and Pentagon 2.

Multiple faults may be suspected when fault identifications provided by Triangles 1, 4, 5 and Pentagons 1, 2 for the same DGA results are different. This is because each graphical representation is more sensitive to some gases and some faults than to others. For example Triangle 4 and the Pentagons are more sensitive to hydrogen and faults S and PD, while Triangles 1 and 5 are more sensitive to ethylene and faults T3.

If the position of the DGA point changes with time in the Triangles and the Pentagons, this indicates that a new fault has formed over the old one. To get a better identification of this new fault, gas concentrations from the previous DGA results may be subtracted from the most recent ones.

The subtracted (delta) values will thus be due only to the new fault. If delta values are negative for some gases, this means that since the previous sample no additional amounts (zero ppm) of these gases have been formed, and that some of those previously formed have started to escape from the transformer. For identifying the new fault, negative delta values should therefore be replaced by zero ppm.

The possible presence of multiple faults may be useful information during the inspection of transformers to avoid having to re-inspect them.

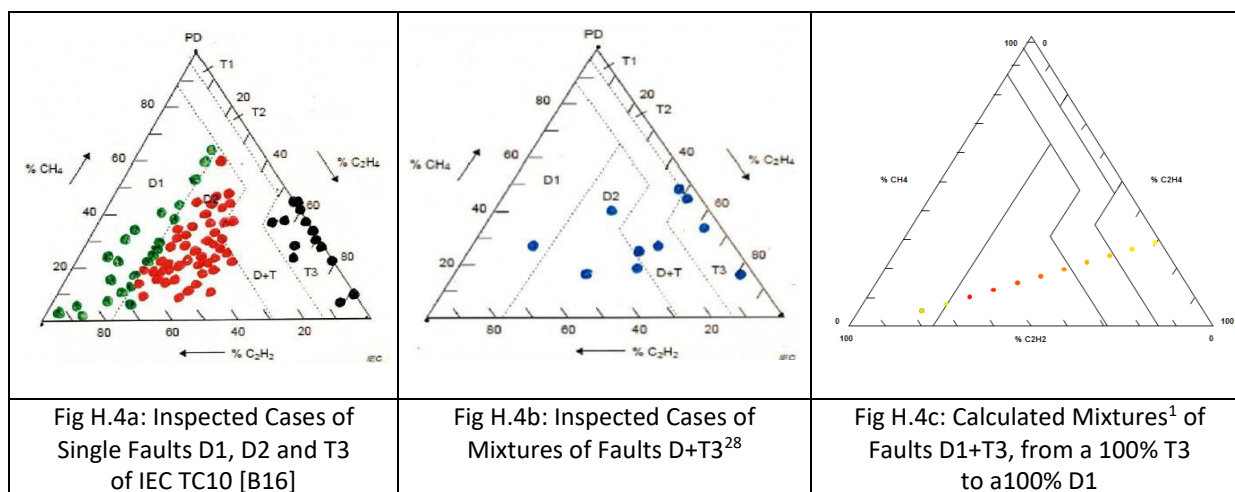
H.7. Effect of multiple faults on dga interpretation methods

App Figure H.4a shows inspected cases of single faults D1, D2 and T3 surveyed by IEC TC10 [B35] as they appear in Triangle 1 (green, red and black dots, respectively).

App Figure H.4b shows inspected cases²⁸ of mixtures of faults D+T3 (blue dots), which may inappropriately appear in zone D2 of Triangle 1 when the contribution of faults T3 and D1 in the mixture of faults is similar (e.g., 50%).

However, if the mixture is mostly due to a T3 and only to a little bit of D1, it will appear in zone T3 of Triangle 1. Reversely, if the mixture is mostly due to a D1, it will appear in zone D1. Regardless of which fault is dominant, or of the extreme dominance of one fault in mixtures of faults, it is better to notice any possible existent fault²⁸.

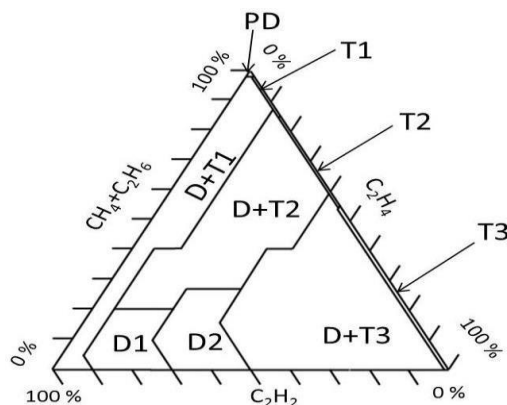
App Figure H.4c shows calculated mixtures¹ of faults T3+D1 as they appear in Triangle 1, starting from a 100% fault T3 (yellow dot) up to a 100% D1 (greenish dot), illustrating that mixtures of faults D+T can appear in any part of the Triangles, Pentagons and other diagnosis methods (e.g., Rogers), depending on the relative proportion of faults D and T in the mixture.



App Figure H.4. Mixtures of Faults vs. Single Faults in Duval Triangle 1

A modified Triangle 1 has been proposed [B9], see App Figure H.5, with most of zones T3, D2 and D1 of the original Duval Triangle 1 replaced by zones of mixtures of faults (D + T). This modified Triangle 1, however, misidentifies as mixtures of faults most of the actual cases of single faults D2, T3 and D1 of IEC indicated in App Figure H.4a. Actual single faults D2 of App Figure H.4a, for example. May be misinterpreted in the modified Triangle 1 as less alarming faults (D+T2). And actual single faults T3 such as in App Figure H.4a (black dots) may send false alarms of discharge faults D in the modified Triangle 1.

Indeed, it has been reported in laboratory pyrolysis tests that acetylene starts forming in oil at temperatures > 650°C in the absence of any arcing discharge¹.

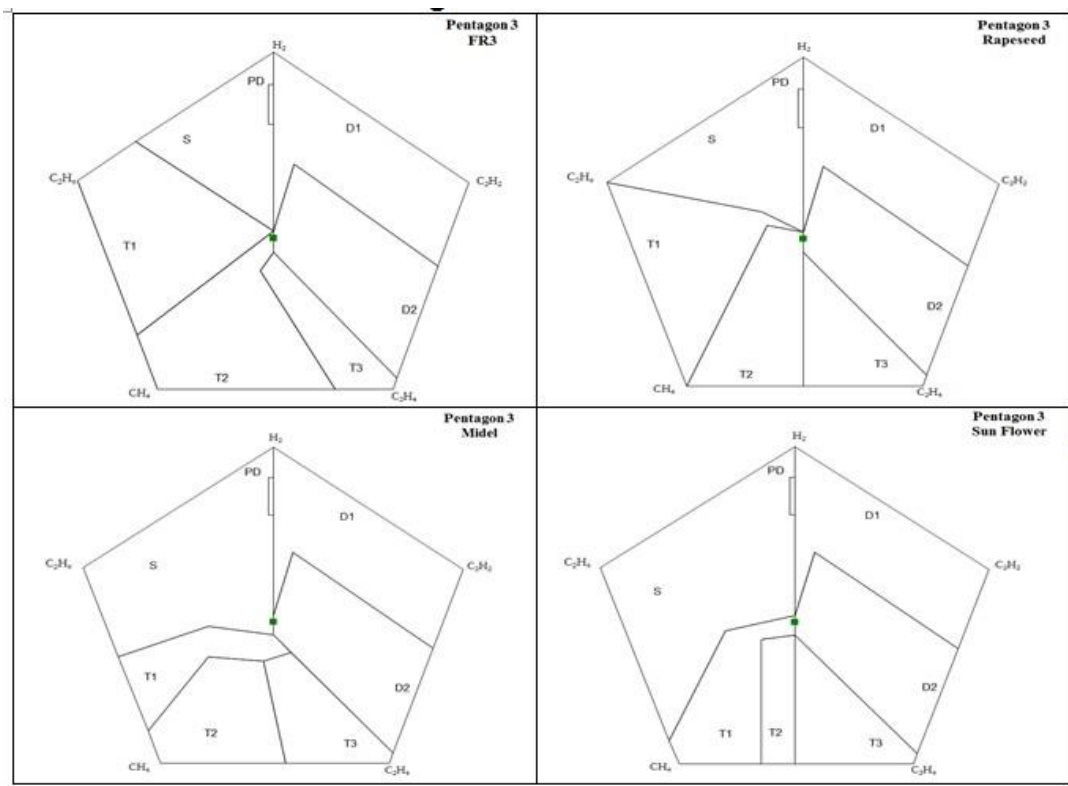


App Figure H.5. Modified Duval Triangle 1 of Spremic [B36]

H.8. Duval triangles 3 and pentagons 3 for non-mineral oils

Natural and synthetic esters are widely used in distribution transformers, and increasingly so in power transformers, because they are less-flammable and more environment-friendly in case of spills, despite some limitations (e.g., higher cost, required protection from oil oxidation, very-low temperature operation, full clean-up required by most national and regional regulatory bodies in case of spills).

Duval Triangles 3, 6 and 7 for non-mineral oils (natural and synthetic esters and silicones) are described in TB 443 [B2]. Since then, Duval Pentagons 3 for non-mineral oils have been developed and are indicated in App Figure H.6.



App Figure H.6. Duval Pentagons 3 for Non-Mineral Oils

Note: for numerical values of zone boundaries, see [B 40] .

APPENDIX I. Fault identification methods using C₃ hydrocarbon gases

I.1. Fault identification methods for transformers

Fault identification methods using C₃ hydrocarbons in Poland are indicated in App Table I.2. Additional ratios C₃H₆/C₃H₈ and C₂H₄/C₃H₈ are used mainly to confirm the temperature range in thermal faults. When ratios are in different ranges the temperature is estimated as close to the border.

Those used in Germany are indicated in App Table I.3 (simplified MSS version) and App Table I.1 (full MSS version).

App Table I.2. Diagnosis Method for C₃ Hydrocarbons in Poland¹⁹

Source	Ratios of characteristic gases	Values of hydrocarbons ratios and temperature ranges		
		(150÷300)°C	(300÷700)°C	> 700°C
Acc. to IEC 60599	C ₂ H ₄ /C ₂ H ₆	< 1	1 ÷ 4	> 4
Complementary developed in <u>Energopomiar</u>	C ₃ H ₆ /C ₃ H ₈	< 2	2 ÷ 6	> 6
	C ₂ H ₄ /C ₃ H ₈	< 3	3 ÷ 15	> 15

App Table I.3. MSS Diagnosis Method for C₃ Hydrocarbons (simplified version ²⁰)

C ₂ H ₄ /C ₃ H ₆	Fault
0.3 to 1.0	<300°C
1-3	300-1000°C
>3	>1000°C

App Table I.4. MSS Diagnosis Method for C₃ Hydrocarbons (full version ²⁰)

Ratio ranges	Ratio numbers				
	C ₂ H ₂ /C ₂ H ₄	H ₂ /CH ₄	C ₂ H ₄ /C ₂ H ₆	C ₂ H ₄ /C ₃ H ₆	CO ₂ /CO
< 0.3	0	0	0	0	1
0.3 to < 1.0	1	0	0	1	1
1.0 to < 3.0	1	1	1	2	1
3.0 to < 10.0	2	2	1	3	0
> 10.0	2	3	1	3	2
Diagnosis	Number sequences				
Normal ageing of insulants	0	0	0	0	0
Discharge of high energy	2	1	1	2/3	1
Discharge of low energy	2	2	1	2/3	1
Partial discharges with high energy	1	3	0	-	0
Partial discharges with low energy	0	3	0	-	0
Local overheating up to 300°C	0	0	0	1	2
Local overheating from 300 to 1000°C	0	0	1	2	2
Local overheating over 1000°C	1	0	1	2/3	2
Local overheating and discharge	1	1	1	2	2
Local overheating and partial discharge	0	3	1	2	2

Note: C₃ hydrocarbons have been used as memory tracers due to their high solubilities, e.g. after a selector fault that has been repaired²⁰.

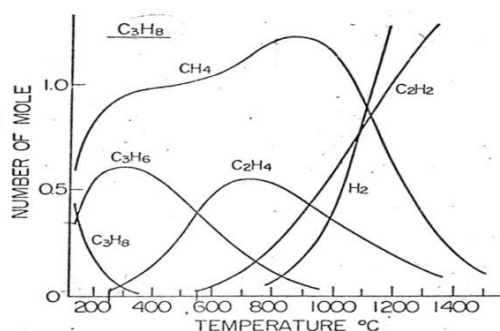
The WG has investigated these various interpretation methods using the C_3 hydrocarbon gases, and the conclusion is that C_3 hydrocarbon gases confirm the diagnosis given by C_1/C_2 hydrocarbon gases but do not really provide additional information.

The DGA database of the WG has been used to calculate the 90% percentile (typical) values of propylene and propane, when these values were reported in the individual databases¹⁴. They are relatively similar to those of C_1/C_2 hydrocarbons:

- 120 ppm for propane vs 80 ppm for methane, ethane.
- 30 ppm for propylene vs 55 ppm for ethylene.

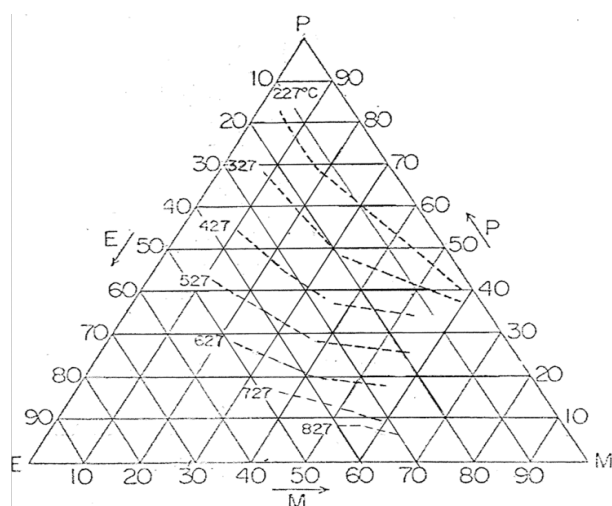
I.2. PEM method for transformers

Another fault identification method using the C_3 hydrocarbon gases, the PEM method, was developed in 1979 in Thailand and Japan [B37], and correlates the theoretical thermodynamic formation of C_3 hydrocarbon gases to oil temperatures from 220°C to 820°C (App Figure I.1).



App Figure I.1. Theoretical Thermodynamic Formation of C_3 Hydrocarbon Gases vs. Temperature

The PEM (Propylene, Ethylene, Methane) Method correlates the relative percentage of propylene, ethylene and methane, with the theoretical thermodynamic values of temperature of oil, from 220°C to 820°C in App Figure I.2. These theoretical values may be significantly different from actual values of oil temperature in transformers, however, so they should be used only with caution.



App Figure I.2. The P.E.M. method

Abnormality is suspected when PEM temperature is $> 300^\circ\text{C}$. The transformer is removed for inspection when it is $> 700^\circ\text{C}$, especially when there is some acetylene.

The following example of gas formation in a transformer has been provided² to illustrate the PEM Method and is indicated in App Table I.5.

App Table I.5. Fault Identification Using C₃ Hydrocarbon Gases

Date	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	C ₃ H ₆	C ₃ H ₈	CO	CO ₂	PEM Temp	Full MSS	Simpl MSS	Polish Temp
09/2014	551	187	152	351	66	148	27	113	2870	677	DT	>300°C	~700°C
01/2015	873	367	109	663	137	270	46	194	2448	727	N/D	>300°C	~700°C
02/2015	1363	572	167	896	142	323	53	314	5321	777	N/D	>300°C	>700°C

N/D =Not Determined

I.3. PEM method for on-load tap-changers

EGAT (Thailand) shares their experience in the estimation of overheating temperature of faulty on-load tap-changers using the PEM method [B37].

The percentage of each gas to the total concentration of these three gases is calculated to plot a dot in the PEM Triangle graph (App Figure I.2). The temperature is estimated by the dotted temperature lines inside the PEM Triangle.

I.3.1. Conventional type on-load tap-changer in normal operation, with acetylene > ethylene

Hydrogen, acetylene and ethylene are significant in normal operation of conventional type on-load tap-changers. In this case of normal arcing, acetylene > ethylene and the estimated temperature from P.E.M. Method is > 727°C. Case studies in case of normal operation are shown in App Table I.6, with P.E.M. value of each DGA result shown in the last column.

App Table I.6. Case Studies of Conventional Type On-Load Tap-Changer in Normal Operation

Name	O ₂	N ₂	CO ₂	CO	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	C ₃ H ₆	C ₃ H ₈	TCG	Code	PEM
S-KT2AB	4550	44290	2292	231	13359	4542	13975	13396	1943	1763	133	49342	102	>827
T-KT1AA	15660	39180	1965	268	20985	4491	35573	9082	1396	2132	77	74004	202	>757

I.3.2. Conventional type on-load tap-changers in abnormal operation, with acetylene > ethylene

If acetylene > ethylene but the estimated temperature from P.E.M. Method is < 727°C as in App Table I.7, on-load tap-changer is in abnormal condition (the first and the last data in App Table I.7).

When the normal condition of data “I” in App Table I.7 became the abnormal condition of data “J”, overheating (resistor burnt & bad contact) was found during internal inspection. Noticed here for data “J” is also a significantly increase of carbon monoxide.

App Table I.7. Case Studies of Conventional Type On-Load Tap-Changers in Abnormal Operation

Name	O ₂	N ₂	CO ₂	CO	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	C ₃ H ₆	C ₃ H ₈	TCG	Code	PEM
H-KT3A	19609	52961	1123	84	227	122	4941	241	7	28	0	5650	122	527
I	20401	66299	781	346	12615	1672	8364	1731	71	285	0	25084	202	770
J	8504	37556	1704	5616	22859	9427	38747	13922	876	3611	83	95141	102	727
L	0	0	417	8	7	197	1238	882	60	461	0	2853	122	660

In App Table I.8 what was found during the inspection of conventional type on-load tap-changer is presented together with the P.E.M. value of each case.

App Table I.8. PEM Value and What was Found in Each Case

Name	Inspection	Code	PEM
H-KT3A	Overheat	122	527
I	Main contact of LTC arc discharge	202	770
J	Resistor burn and bad contact	102	727
L	LTC arc	122	660

I.3.3. Overheating in conventional type on-load tap-changer, with ethylene > acetylene

In case of ethylene > acetylene, overheating is usually suspected. DGA results of other phases in the same transformer or DGA results of other transformer(s) with the same manufacturer and same model are used for comparison.

I.3.4. Conventional type on-load tap-changer in normal operation, with ethylene > acetylene

If all three phases of the on-load tap-changer have ethylene > acetylene, and the temperature from P.E.M. Method is > 827°C, the on-load tap-changers are in normal operation as in the case studies in App Table I.9. (In the past, EGAT always identified overheating by ethylene > acetylene, but there were many cases where no faults were found from internal inspection).

App Table I.9. Case Studies of Conventional Type On-Load Tap-Changers in Normal Condition in spite of Ethylene > Acetylene

Name	O ₂	N ₂	CO ₂	CO	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	C ₃ H ₆	C ₃ H ₈	TCG	Code	PEM
M-KT3AA	7241	65026	44	48	12690	2325	220	974	264	0	16	16537	102	>827
M-KT3AB	7357	63111	57	105	9481	1952	303	812	187	0	7	12847	102	>827
M-KT3AC	8562	62284	69	30	8889	2269	207	887	229	76	29	12616	102	>827

I.3.5. P.E.M. method for vacuum type on-load tap-changers

For vacuum type on-load tap-changers, the DGA interpretation is more or less similar to the interpretation of faults in the transformer main tank.

CIGRE

21, rue d'Artois
75008 Paris - FRANCE



© CIGRE

ISBN : 978-2-85873-473-3